



CS 412 Intro. to Data Mining

Chapter 8. Classification: Basic Concepts

Jiawei Han, Computer Science, Univ. Illinois at Urbana-Champaign, 2017





Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts 
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary

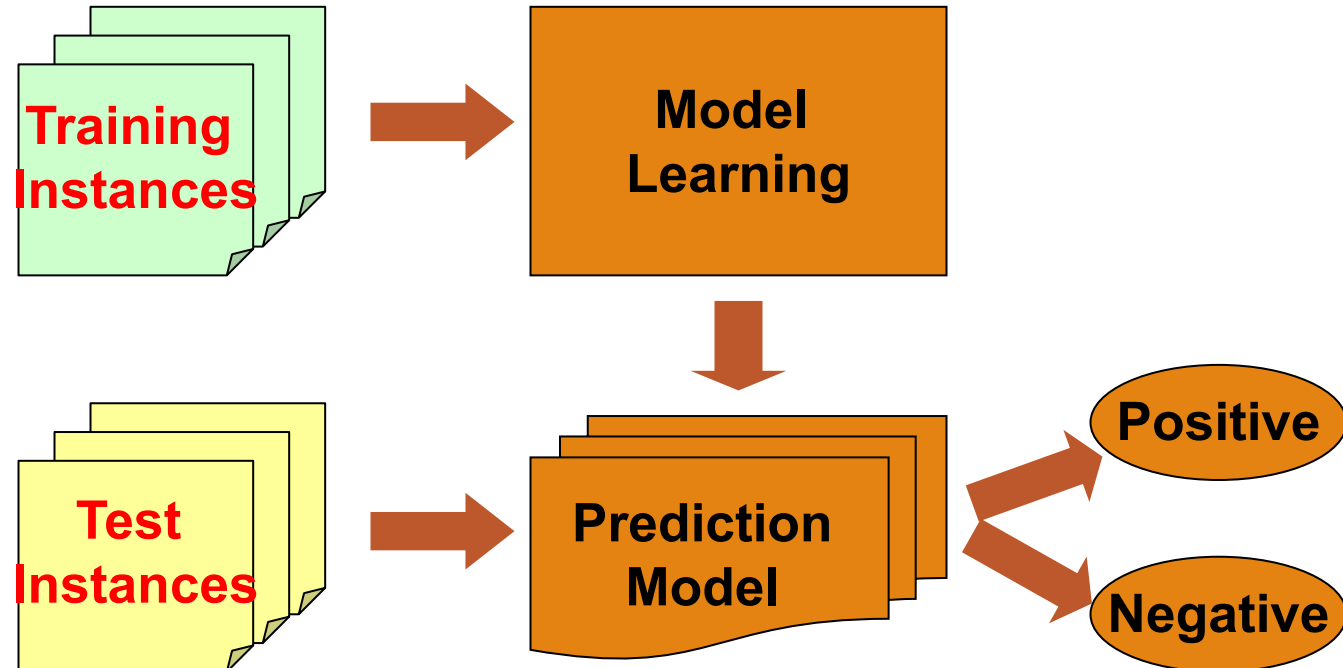
Supervised vs. Unsupervised Learning (1)

□ Supervised learning (classification) *à lable*

- Supervision: The training data such as observations or measurements are accompanied by **labels** indicating the classes which they belong to
- New data is classified based on the models built from the training set

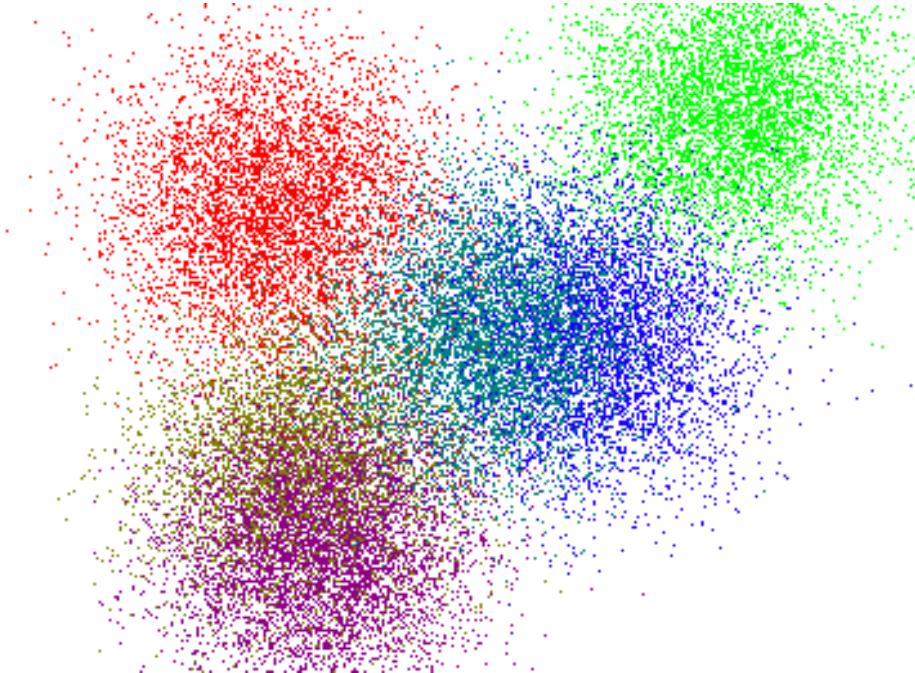
Training Data with class label:

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



Supervised vs. Unsupervised Learning (2)

- Unsupervised learning (clustering) *بدون تسمية*
 - The class labels of training data are unknown
 - Given a set of observations or measurements, establish the possible existence of classes or clusters in the data



Prediction Problems: Classification vs. Numeric Prediction

□ Classification

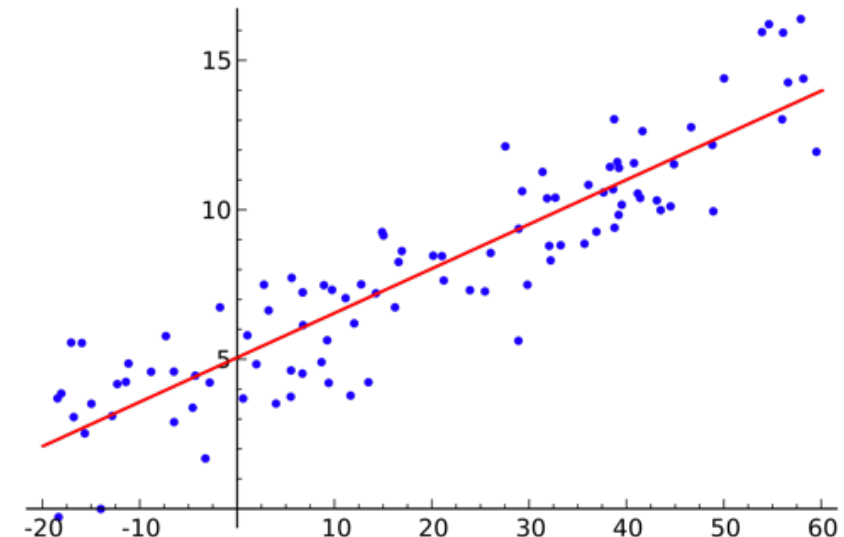
- Predict categorical class labels (discrete or nominal)
- Construct a model based on the training set and the **class labels** (the values in a classifying attribute) and use it in classifying new data

□ Numeric prediction

- Model continuous-valued functions (i.e., predict unknown or missing values)

□ Typical applications of classification

- Credit/loan approval
- Medical diagnosis: if a tumor is cancerous or benign
- Fraud detection: if a transaction is fraudulent
- Web page categorization: which category it is



Classification—Model Construction, Validation and Testing

❑ Model construction

- ❑ Each sample is assumed to belong to a predefined class (shown by the **class label**)
- ❑ The set of samples used for model construction is **training set**
- ❑ Model: Represented as decision trees, rules, mathematical formulas, or other forms

❑ Model Validation and Testing:

- ❑ **Test:** Estimate accuracy of the model
 - ❑ The known label of test sample is compared with the classified result from the model
 - ❑ *Accuracy:* % of test set samples that are correctly classified by the model
 - ❑ Test set is independent of training set
- ❑ **Validation:** If *the test set* is used to select or refine models, it is called **validation** (or development) **(test) set**

❑ Model Deployment: If the accuracy is acceptable, use the model to classify new data

Chapter 8. Classification: Basic Concepts

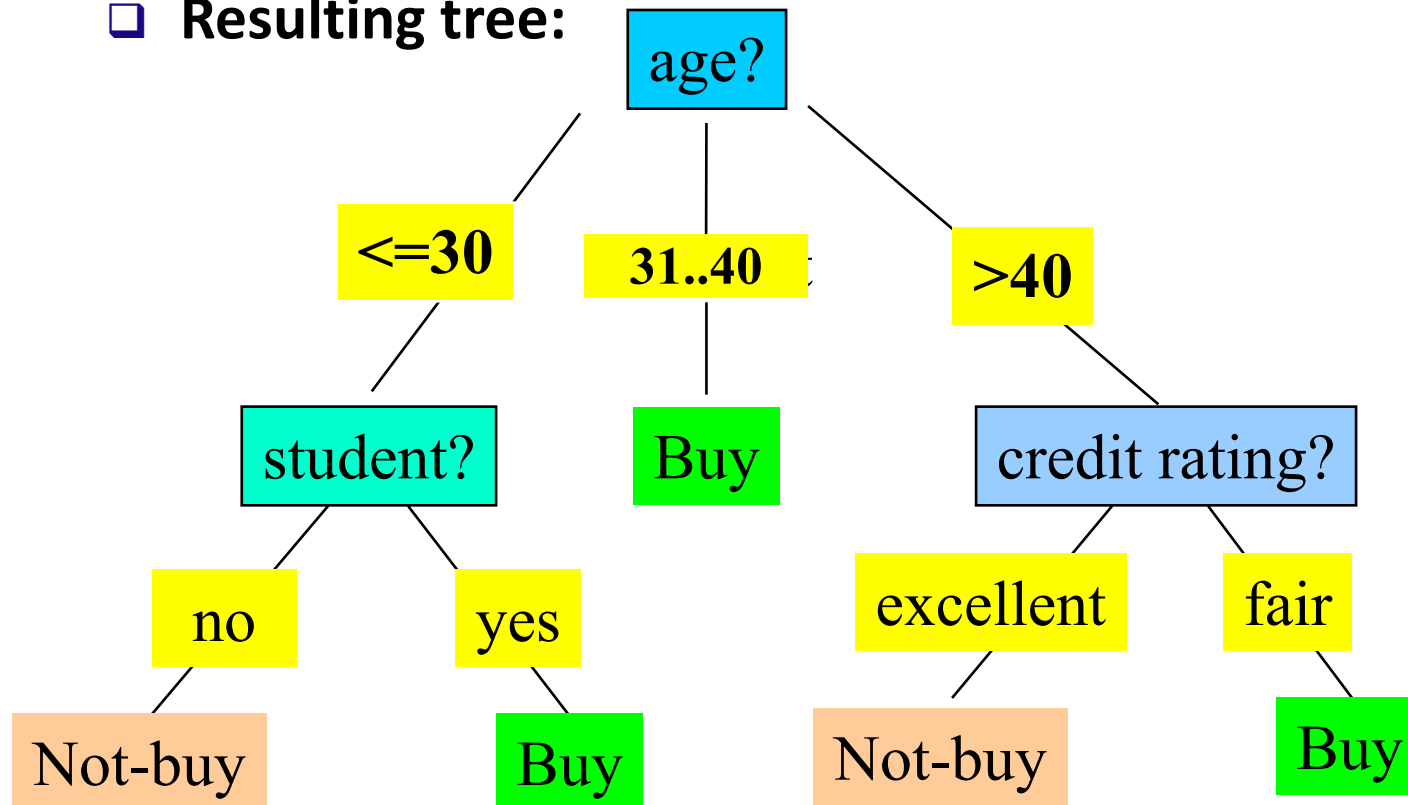
- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction 
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary

Decision Tree Induction: An Example

Decision tree construction:

- A top-down, recursive, divide-and-conquer process

Resulting tree:



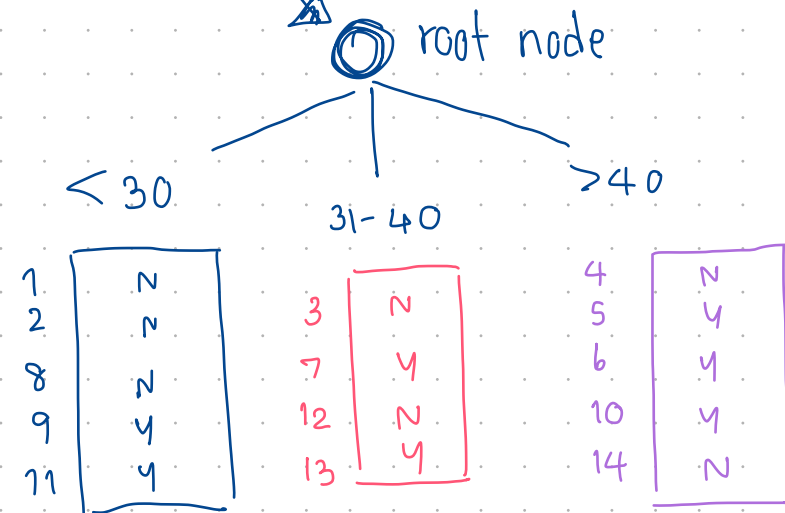
Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from
"Playing Tennis" example of R. Quinlan

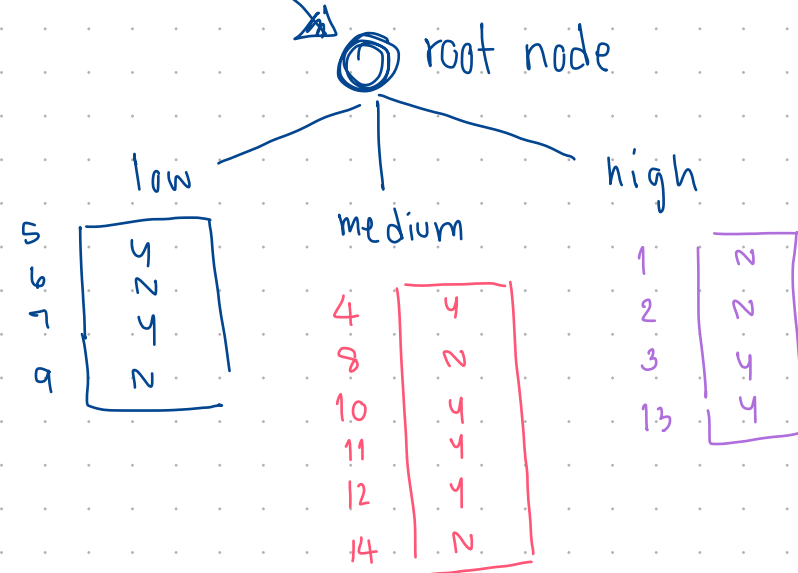
Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no

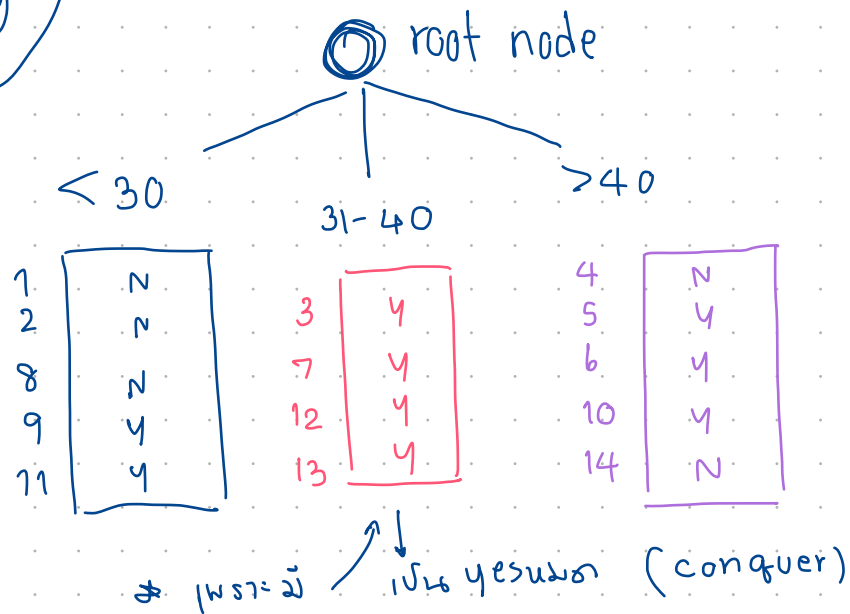


Training data set: Who buys computer?

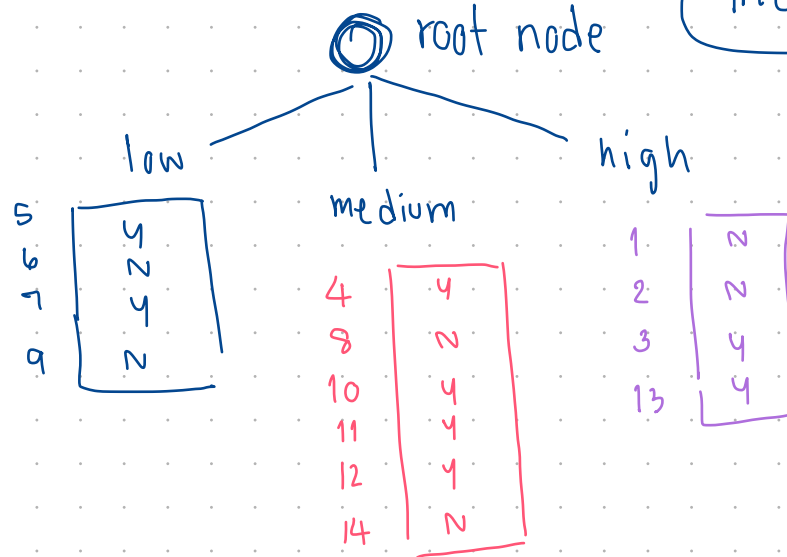
	age	income	student	credit_rating	buys_computer
	<=30	high	no	fair	no
	<=30	high	no	excellent	no
4	31...40	high	no	fair	yes
5	>40	medium	no	fair	yes
6	>40	low	yes	fair	yes
7	>40	low	yes	excellent	no
8	31...40	low	yes	excellent	yes
9	<=30	medium	no	fair	no
10	<=30	low	yes	fair	yes
11	>40	medium	yes	fair	yes
12	<=30	medium	yes	excellent	yes
13	31...40	medium	no	excellent	yes
14	31...40	high	yes	fair	yes
15	>40	medium	no	excellent	no



age

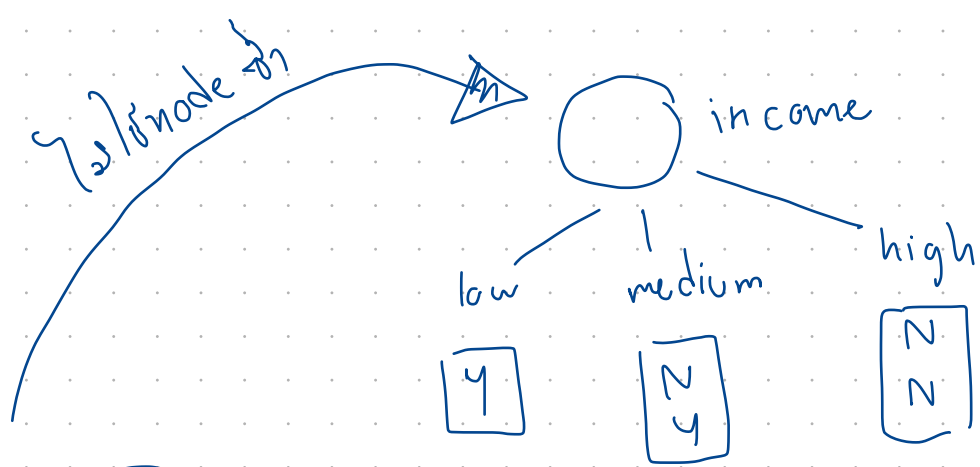


income

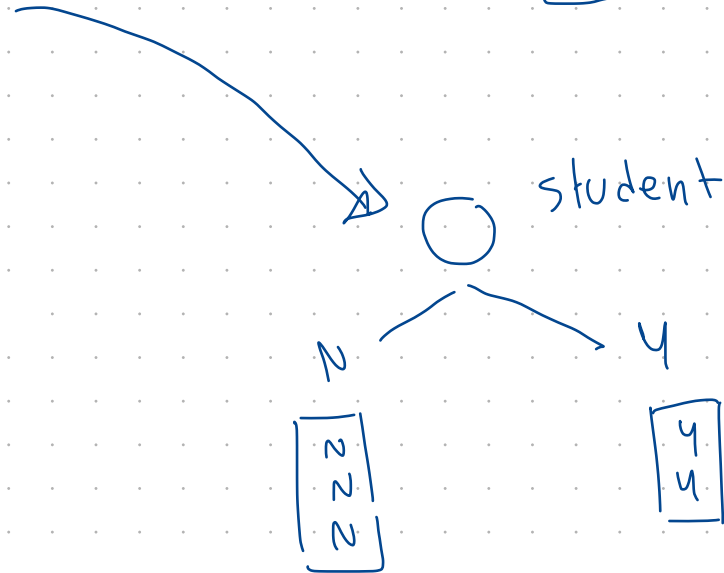


1
2
8
9
11

2
2
2
2
2

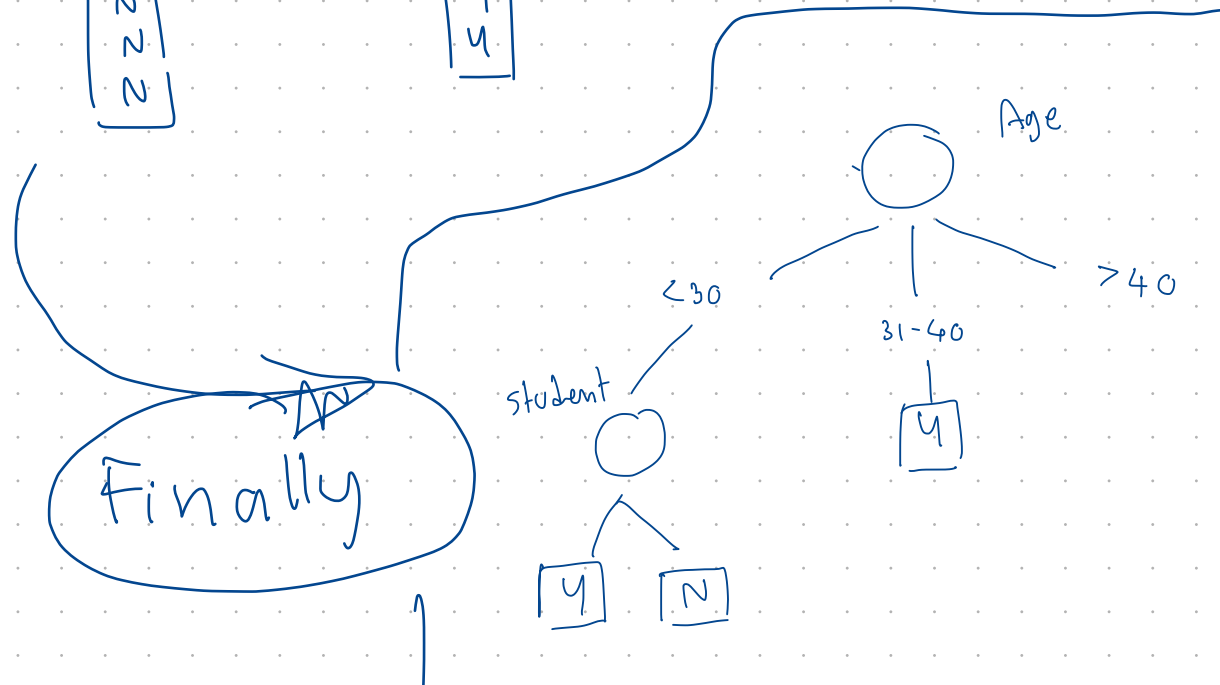


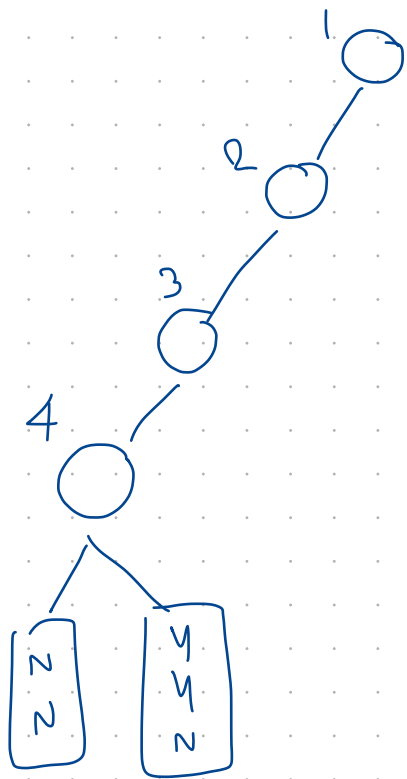
age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes



Training data set: Who buys computer?

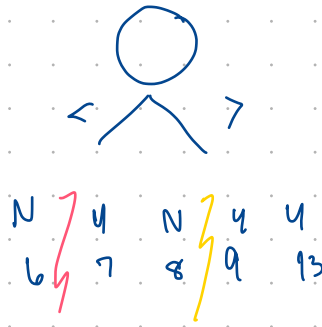
age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no





majority vote
 $y \rightarrow 66\% = 4$

P	
13	4
6	2
7	4
8	2
9	4



* not continuous best splitting point

Quiz 9 info 7 nu 7

$$\text{Info}(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$= - I(3, 2)$$

$$= - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right)$$

$$= 0.97$$

$$\text{Gain} = 0.97 - 0.6488$$

$$= 0.32$$

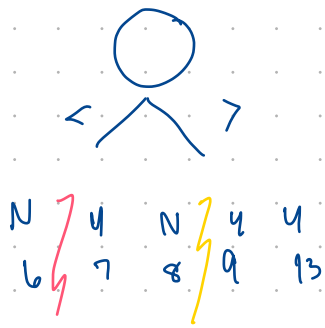
$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{info}(D_j)$$

$$= \frac{1}{5} I(0, 1) + \frac{4}{5} I(3, 1)$$

$$= \frac{1}{5} (0) + \frac{4}{5} (0.311 + 0.5)$$

$$= 0.6488$$

$$\frac{3}{4} \log_2 \left(\frac{3}{4} \right) + \frac{1}{4} \log_2 \left(\frac{1}{4} \right)$$



Best splitting point

Quiz 9 info 7 nu 7

$$\text{Info}(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$= -I(3, 2)$$

$$= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right)$$

$$= 0.97$$

$$\text{Gain} = 0.97 - 0.6488$$

$$= 0.32$$

$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{info}(D_j)$$

$$= \frac{1}{5} I(0, 1) + \frac{4}{5} I(3, 1)$$

$$= \frac{1}{5} (0) + \frac{4}{5} (0.311 + 0.5)$$

$$= 0.6488$$

$$\text{Info}_A(D) = \frac{3}{5} I(1, 2) + \frac{2}{5} I(2, 0)$$

$$= \frac{3}{5} (0.53 + 0.19) + \frac{2}{5} (0)$$

$$= 0.552$$

$$\text{Gain} = 0.97 - 0.552$$

$$= 0.418$$

From Entropy to Info Gain: A Brief Review of Entropy

□ Entropy (Information Theory)

□ A measure of uncertainty associated with a random number

□ Calculation: For a discrete random variable Y taking m distinct values $\{y_1, y_2, \dots, y_m\}$

$$H(Y) = - \sum_{i=1}^m p_i \log(p_i) \quad \text{where } p_i = P(Y = y_i)$$

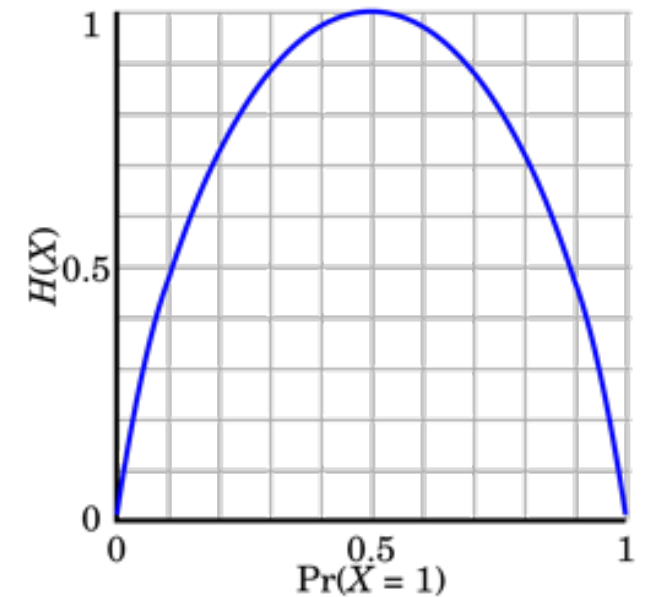
□ Interpretation

□ Higher entropy $\rightarrow$ higher uncertainty

□ Lower entropy $\rightarrow$ lower uncertainty

□ Conditional entropy

$$H(Y|X) = \sum_x p(x) H(Y|X = x)$$



m = 2

Information Gain: An Attribute Selection Measure

- ❑ Select the attribute with the highest information gain (used in typical decision tree induction algorithm: ID3/C4.5)
- ❑ Let p_i be the probability that an arbitrary tuple in D belongs to class C_i , estimated by $|C_{i,D}|/|D|$
- ❑ Expected information (entropy) needed to classify a tuple in D :

info(D)
$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$
 ก. ค่าจ. info ที่จ. = 0.5054 + ก. ค่าจ. info ที่จ. = 0.5054

- ❑ Information needed (after using A to split D into v partitions) to classify D :

info_A(D)
$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$
 ก. Data ที่จ. และ ก. Node ที่จ. และ ก. Data ที่จ. และ ก. Node

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times info(D_j)$$

- ❑ Information gained by branching on attribute A

$$Gain(A) = \underset{\text{info(D)}}{Info(D)} - \underset{\text{info}_A(D)}{Info_A(D)}$$

Example: Attribute Selection with Information Gain

□ Class P: buys_computer = “yes”

□ Class N: buys_computer = “no”

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

age	p _i	n _i	I(p _i , n _i)
<=30	2	3	0.971
31...40	4	0	0
>40	3	2	0.971

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) = 0.694$$

$-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4}$

$\frac{5}{14} I(2,3)$ means “age <=30” has 5 out of 14 samples, with 2 yes’es and 3 no’s.

Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

Similarly, we can get

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit_rating) = 0.048$$

Decision Tree Induction: Algorithm

- ❑ Basic algorithm
 - ❑ Tree is constructed in a **top-down, recursive, divide-and-conquer manner**
 - ❑ At start, all the training examples are at the root
 - ❑ Examples are partitioned recursively based on selected attributes
 - ❑ On each node, attributes are selected based on the training examples on that node, and a heuristic or statistical measure (e.g., **information gain**)
- ❑ Conditions for stopping partitioning
 - ❑ All samples for a given node belong to the same class
 - ❑ There are no remaining attributes for further partitioning
 - ❑ There are no samples left
- ❑ Prediction
 - ❑ **Majority voting** is employed for classifying the leaf

Quiz 9 info

How to Handle Continuous-Valued Attributes?

- ❑ Method 1: Discretize continuous values and treat them as categorical values
 - ❑ E.g., age: < 20, 20..30, 30..40, 40..50, > 50
- ❑ Method 2: Determine the **best split point** for continuous-valued attribute A
 - ❑ Sort the value A in increasing order:, e.g. 15, 18, 21, 22, 24, 25, 29, 31, ...
 - ❑ *Possible split point*: the midpoint between *each pair of adjacent values*
 - ❑ $(a_i + a_{i+1})/2$ is the midpoint between the values of a_i and a_{i+1}
 - ❑ e.g., $(15+18)/2 = 16.5, 19.5, 21.5, 23, 24.5, 27, 30, \dots$
 - ❑ The point with the *maximum information gain* for A is selected as the **split-point** for A
- ❑ Split: Based on split point P
 - ❑ The set of tuples in D satisfying $A \leq P$ vs. those with $A > P$

Gain Ratio: A Refined Measure for Attribute Selection

- ❑ Information gain measure is biased towards attributes with a large number of values
- ❑ Gain ratio: Overcomes the problem (as a normalization to information gain)

$$\text{SplitInfo}_A(D) = -\sum_{j=1}^v \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right)$$

- ❑ $\text{GainRatio}(A) = \text{Gain}(A) / \text{SplitInfo}(A)$
- ❑ The attribute with the maximum gain ratio is selected as the splitting attribute
- ❑ Gain ratio is used in a popular algorithm C4.5 (a successor of ID3) by R. Quinlan
- ❑ Example
 - ❑ $\text{SplitInfo}_{\text{income}}(D) = -\frac{4}{14} \log_2 \frac{4}{14} - \frac{6}{14} \log_2 \frac{6}{14} - \frac{4}{14} \log_2 \frac{4}{14} = 1.557$
 - ❑ $\text{GainRatio}(\text{income}) = 0.029 / 1.557 = 0.019$

Another Measure: Gini Index

- Gini index: Used in CART, and also in IBM IntelligentMiner
- If a data set D contains examples from n classes, gini index, $gini(D)$ is defined as
 - $gini(D) = 1 - \sum_{j=1}^n p_j^2$
 - p_j is the relative frequency of class j in D
- If a data set D is split on A into two subsets D_1 and D_2 , the $gini$ index $gini(D)$ is defined as
 - $gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$
- Reduction in Impurity:
 - $\Delta gini(A) = gini(D) - gini_A(D)$
- The attribute provides the smallest $gini_{split}(D)$ (or the largest reduction in impurity) is chosen to split the node (*need to enumerate all the possible splitting points for each attribute*)

Computation of Gini Index

- Example: D has 9 tuples in buys_computer = “yes” and 5 in “no”

$$gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

- Suppose the attribute income partitions D into 10 in D_1 : {low, medium} and 4 in D_2

- $$\begin{aligned} gini_{income \in \{low, medium\}}(D) &= \frac{10}{14} gini(D_1) + \frac{4}{14} gini(D_2) \\ &= \frac{10}{14} \left(1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 \right) = 0.443 \\ &= Gini_{income \in \{high\}}(D) \end{aligned}$$

- $Gini_{\{low, high\}}$ is 0.458; $Gini_{\{medium, high\}}$ is 0.450

- Thus, split on the {low,medium} (and {high}) since it has the lowest Gini index

- All attributes are assumed continuous-valued
- May need other tools, e.g., clustering, to get the possible split values
- Can be modified for categorical attributes

Comparing Three Attribute Selection Measures

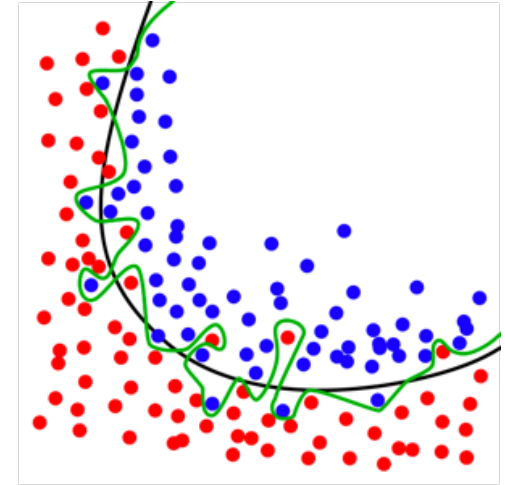
- ❑ The three measures, in general, return good results but
 - ❑ **Information gain:**
 - ❑ biased towards multivalued attributes
 - ❑ **Gain ratio:**
 - ❑ tends to prefer unbalanced splits in which one partition is much smaller than the others
 - ❑ **Gini index:**
 - ❑ biased to multivalued attributes
 - ❑ has difficulty when # of classes is large
 - ❑ tends to favor tests that result in equal-sized partitions and purity in both partitions

Other Attribute Selection Measures

- ❑ Minimal Description Length (MDL) principle
 - ❑ Philosophy: The simplest solution is preferred
 - ❑ The best tree as the one that requires the fewest # of bits to both (1) encode the tree, and (2) encode the exceptions to the tree
- ❑ CHAID: a popular decision tree algorithm, measure based on χ^2 test for independence
- ❑ Multivariate splits (partition based on multiple variable combinations)
 - ❑ CART: finds multivariate splits based on a linear combination of attributes
- ❑ There are many other measures proposed in research and applications
 - ❑ E.g., G-statistics, C-SEP
- ❑ Which attribute selection measure is the best?
 - ❑ Most give good results, none is significantly superior than others

Overfitting and Tree Pruning

- ❑ Overfitting: An induced tree may overfit the training data
 - ❑ Too many branches, some may reflect anomalies due to noise or outliers
 - ❑ Poor accuracy for unseen samples
- ❑ Two approaches to avoid overfitting
 - ❑ Prepruning: *Halt tree construction early*—do not split a node if this would result in the goodness measure falling below a threshold
 - ❑ Difficult to choose an appropriate threshold
 - ❑ Postpruning: *Remove branches* from a “fully grown” tree—get a sequence of progressively pruned trees
 - ❑ Use a set of data different from the training data to decide which is the “best pruned tree”



Classification in Large Databases

- ❑ Classification—a classical problem extensively studied by statisticians and machine learning researchers
- ❑ Scalability: Classifying data sets with millions of examples and hundreds of attributes with reasonable speed
- ❑ Why is decision tree induction popular?
 - ❑ Relatively fast learning speed
 - ❑ Convertible to simple and easy to understand classification rules
 - ❑ Easy to be adapted to database system implementations (e.g., using SQL)
 - ❑ Comparable classification accuracy with other methods
- ❑ **RainForest** (VLDB'98 — Gehrke, Ramakrishnan & Ganti)
 - ❑ Builds an AVC-list (attribute, value, class label)

RainForest: A Scalable Classification Framework

- The criteria that determine the quality of the tree can be computed separately
 - Builds an AVC-list: **AVC (Attribute, Value, Class_label)**
- **AVC-set** (of an attribute X)
 - Projection of training dataset onto the attribute X and class label where counts of individual class label are aggregated

- **AVC-group** (of a node n)

- Set of AVC-sets of all predictor attributes at the node n

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

The Training Data

AVC-set on Age

Age	Buy_Computer	
	yes	no
<=30	2	3
31..40	4	0
>40	3	2

AVC-set on Income

income	Buy_Computer	
	yes	no
high	2	2
medium	4	2
low	3	1

AVC-set on Student

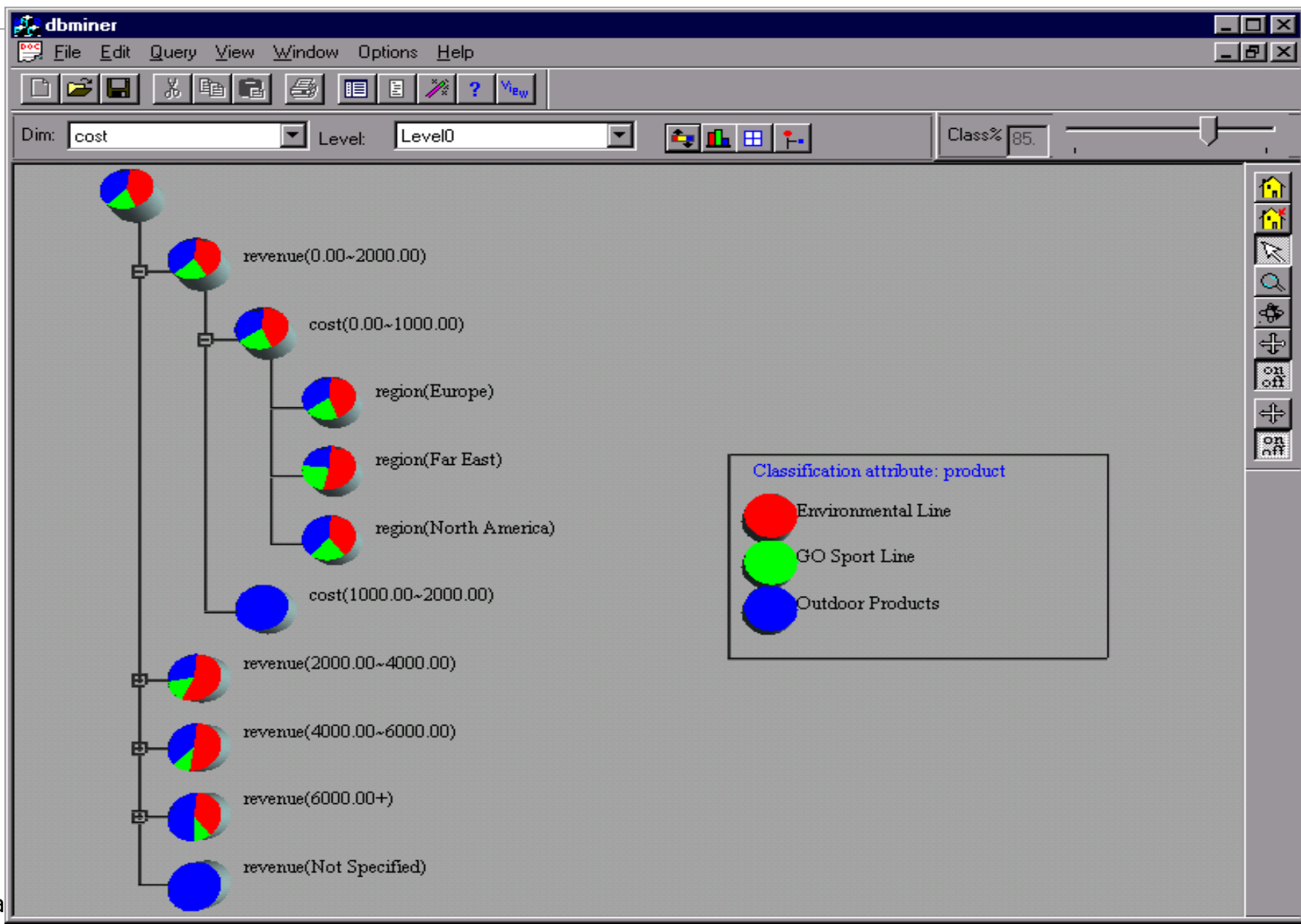
student	Buy_Computer	
	yes	no
yes	6	1
no	3	4

AVC-set on Credit_Rating

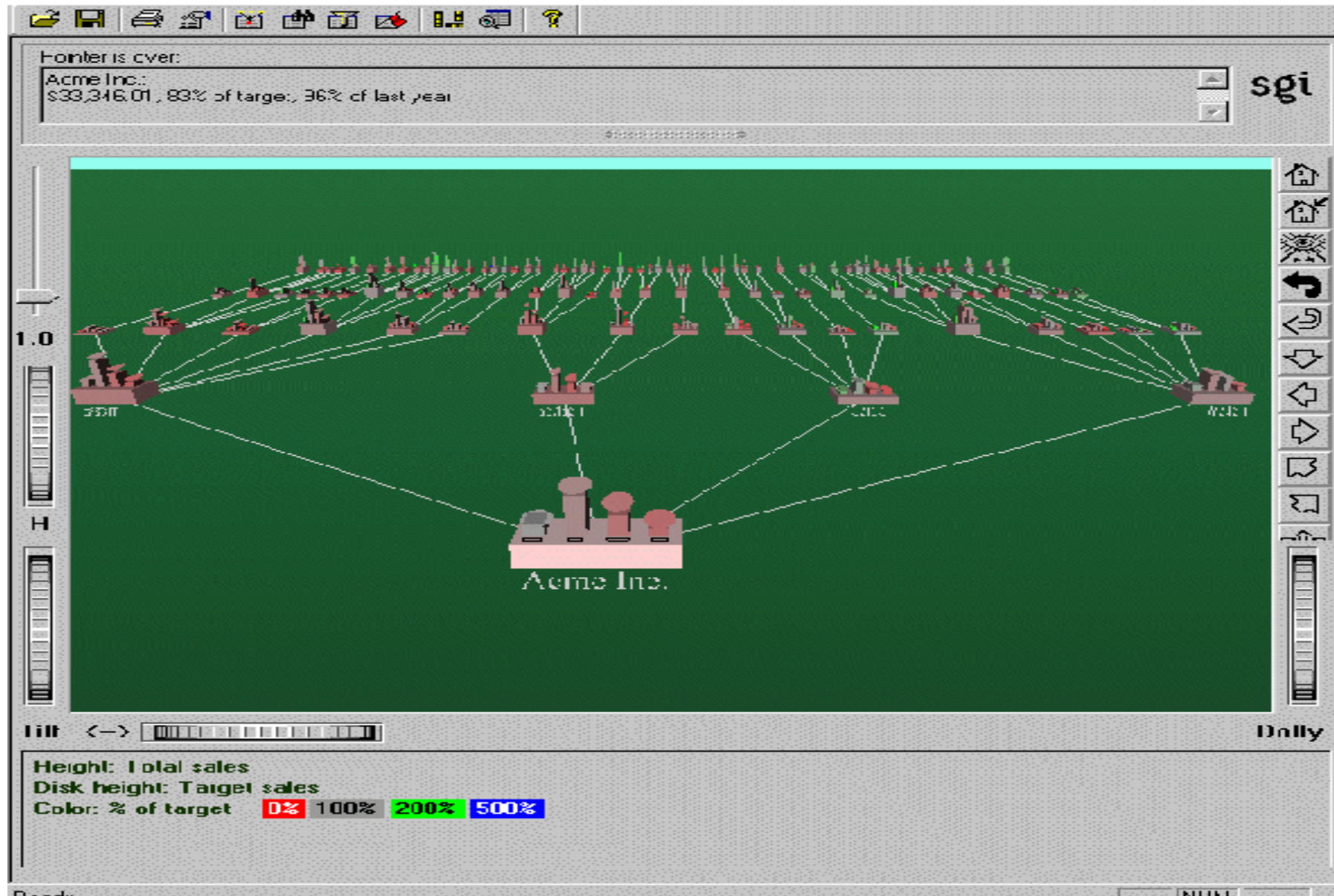
Credit rating	Buy_Computer	
	yes	no
fair	6	2
excellent	3	3

Its AVC Sets

Presentation of Classification Results

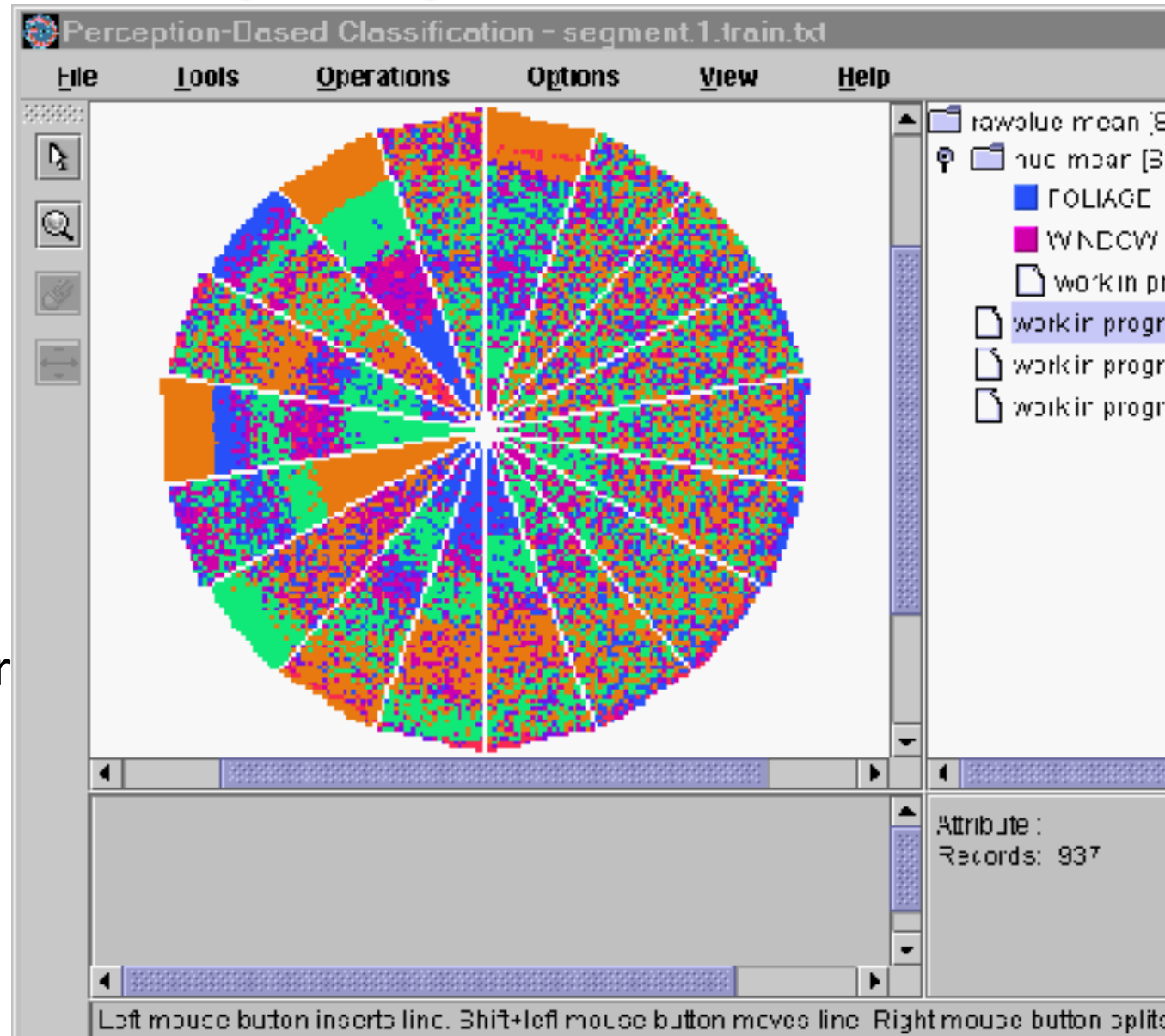


Visualization of a Decision Tree (in SGI/MineSet 3.0)

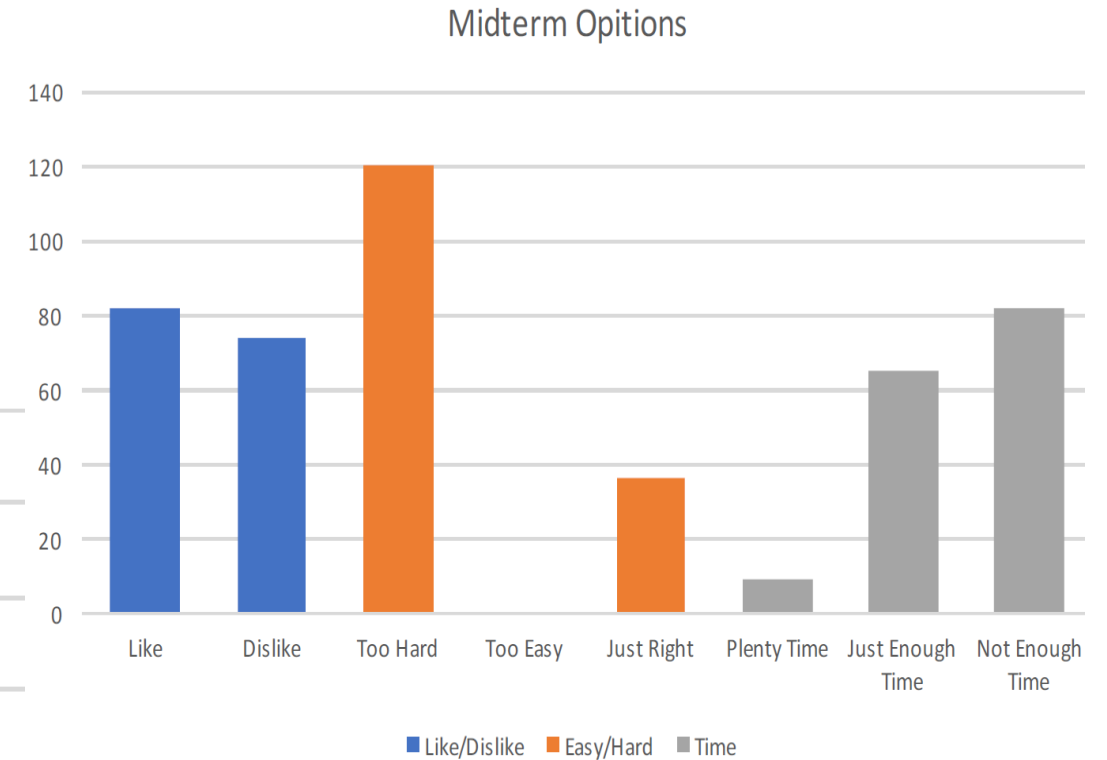
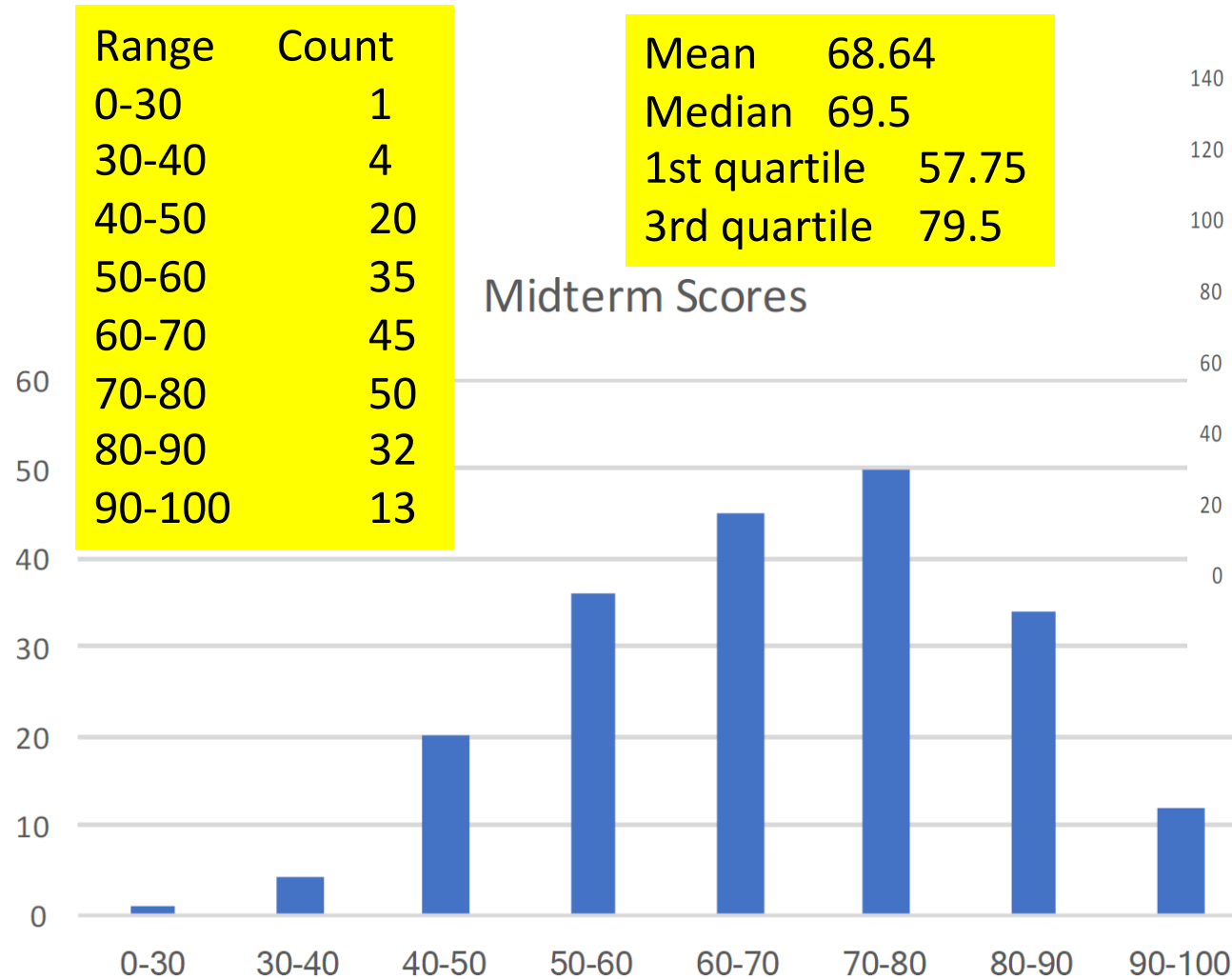


Interactive Visual Mining by Perception-Based Classification (PBC)

- ❑ Perception-based classifier (PCB): developed at Univ. of Munich (1999)
- ❑ One color represents one class label
- ❑ One pie represents one attribute (or variable)
- ❑ The pie with random spread implies weak classification power
- ❑ The pie with clearly partitioned color strips implies good classification power
- ❑ One can select a good attribute and regenerate new pie charts for classification at the subsequent levels



CS412-Fall 2017: Midterm Statistics



Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods 
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary

What Is Bayesian Classification?

- ❑ A statistical classifier
 - ❑ Perform *probabilistic prediction* (i.e., predict class membership probabilities)
- ❑ Foundation—Based on Bayes' Theorem
- ❑ Performance
 - ❑ A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- ❑ Incremental
 - ❑ Each training example can incrementally increase/decrease the probability that a hypothesis is correct—prior knowledge can be combined with observed data
- ❑ Theoretical Standard
 - ❑ Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayes' Theorem: Basics

- Total probability Theorem:

$$p(B) = \sum_i p(B|A_i)p(A_i)$$



- Bayes' Theorem:

$$p(H|X) = \frac{p(X|H)P(H)}{p(X)} \propto p(X|H)P(H)$$

posteriori probability

likelihood

prior probability

What we should choose

What we just see

What we knew previously

- **X**: a data sample ("*evidence*")
- **H**: X belongs to class C

Prediction can be done based on Bayes' Theorem:

Classification is to derive the maximum posteriori

Naïve Bayes Classifier: Making a Naïve Assumption

- ❑ Practical difficulty of Naïve Bayes inference: It requires initial knowledge of many probabilities, which may not be available or involving significant computational cost
- ❑ A Naïve Special Case
 - ❑ Make an additional **assumption** to simplify the model, but achieve comparable performance.

attributes are conditionally independent
(i.e., no dependence relation between attributes)

$$p(X|C_i) = \prod_k p(x_k|C_i) = p(x_1|C_i) \cdot p(x_2|C_i) \cdots p(x_n|C_i)$$

- ❑ Only need to count the class distribution w.r.t. features

X_1	X_2	y
4		4
5		2
2		2

2 9

X_1	X_2	Y
4		4
4		2
2		2

2 ? $\rightarrow$ Test

$$\overbrace{P(H|x)}^{\text{test}} \propto \overbrace{P(x|H) P(H)}^{\text{train}}$$

$$P(y = \text{yes} | \vec{X}) = P(x_1 = n | y = y) P(y = y)$$

Annotations from the image:
 - An arrow points from the $\vec{X}$ in the formula to the text "known yes".
 - An arrow points from the $y = y$ in the formula to the text "No inclusions".
 - The value $\frac{0}{1}$ is written below the $P(x_1 = n | y = y)$ term.
 - The value $\frac{1}{3}$ is written below the $P(y = y)$ term.

Naïve Bayes Classifier: Categorical vs. Continuous Valued Features

၂၅
• x_j ၀ သို့မဟုတ် ၁ ဖြစ်သည်

- If feature x_k is categorical, $p(x_k = v_k | C_i)$ is the # of tuples in C_i with $x_k = v_k$, divided by $|C_{i,D}|$ (# of tuples of C_i in D)

$$p(X|C_i) = \prod_k p(x_k|C_i) = p(x_1|C_i) \cdot p(x_2|C_i) \cdots p(x_n|C_i)$$

- If feature x_k is continuous-valued, $p(x_k = v_k | C_i)$ is usually computed based on Gaussian distribution with a mean μ and standard deviation σ

$$p(x_k = v_k | C_i) = N(x_k | \mu_{C_i}, \sigma_{C_i}) = \frac{1}{\sqrt{2\pi}\sigma_{C_i}} e^{-\frac{(x - \mu_{C_i})^2}{2\sigma^2}}$$

Naïve Bayes Classifier: Training Dataset

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

Data to be classified:

X = (age <=30, Income = medium,
Student = yes, Credit_rating = Fair)

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$P(\text{buy} = y | \hat{X}) = P(\hat{X} | \text{buy} = y) P(\text{buy} = y)$$

$$P(\text{age} \leq 30 | \text{yes}) P(\text{inc} = \text{med} | \text{yes}) P(\text{st} = \text{yes}) P(\text{cf} = \text{cam}, 50)$$

$$\frac{2}{9} \quad \frac{4}{9} \quad \frac{6}{9} \quad \frac{9}{14}$$

$$P(y | \hat{X}) = \frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{9}{14}$$

$$P(N | \hat{X}) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}$$

Quiz 12 ♀, Age = 20, student = yes

$$P(\text{buy} = y | \hat{X}) = P(\hat{X} | \text{buy} = y) P(\text{buy} = y)$$

$$P(\text{age} \leq 30 | \text{yes}) P(\text{st} = \text{yes} | \text{yes})$$

$$\frac{2}{9} \quad \frac{6}{9}$$

$$P(y | \hat{X}) = \frac{2}{9} \times \frac{6}{9} \times \frac{9}{14} = \frac{2}{21} = 0.095$$

$$P(N | \hat{X}) = \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14} = \frac{3}{70} = 0.043$$

Naïve Bayes Classifier: An Example

□ $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

□ Compute $P(X|C_i)$ for each class

$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$

$P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

□ **$X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$**

$P(X|C_i)$: $P(X | \text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$

$P(X | \text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$

$P(X|C_i) * P(C_i)$: $P(X | \text{buys_computer} = \text{"yes"}) * P(\text{buys_computer} = \text{"yes"}) = 0.028$

$P(X | \text{buys_computer} = \text{"no"}) * P(\text{buys_computer} = \text{"no"}) = 0.007$

Therefore, X belongs to class ("buys_computer = yes")

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Avoiding the Zero-Probability Problem

- ❑ Naïve Bayesian prediction requires each conditional probability be **non-zero**

- ❑ Otherwise, the predicted probability will be zero

$$p(X|C_i) = \prod_k p(x_k|C_i) = p(x_1|C_i) \cdot p(x_2|C_i) \cdots p(x_n|C_i)$$

- ❑ Example. Suppose a dataset with 1000 tuples:

income = low (0), income = medium (990), and income = high (10)

- ❑ Use **Laplacian correction** (or Laplacian estimator)

- ❑ *Adding 1 to each case*

$$\text{Prob}(\text{income} = \text{low}) = 1/(1000 + 3)$$

$$\text{Prob}(\text{income} = \text{medium}) = (990 + 1)/(1000 + 3)$$

$$\text{Prob}(\text{income} = \text{high}) = (10 + 1)/(1000 + 3)$$

- ❑ The “corrected” probability estimates are close to their “uncorrected” counterparts

Naïve Bayes Classifier: Strength vs. Weakness

- ❑ Strength
 - ❑ Easy to implement
 - ❑ Good results obtained in most of the cases
- ❑ Weakness
 - ❑ Assumption: attributes conditional independence, therefore loss of accuracy
 - ❑ Practically, dependencies exist among variables
 - ❑ E.g., Patients: Profile: age, family history, etc.
Symptoms: fever, cough etc.
Disease: lung cancer, diabetes, etc.
 - ❑ Dependencies among these cannot be modeled by Naïve Bayes Classifier
- ❑ How to deal with these dependencies?
 - ❑ Use Bayesian Belief Networks (to be covered in the next chapter)

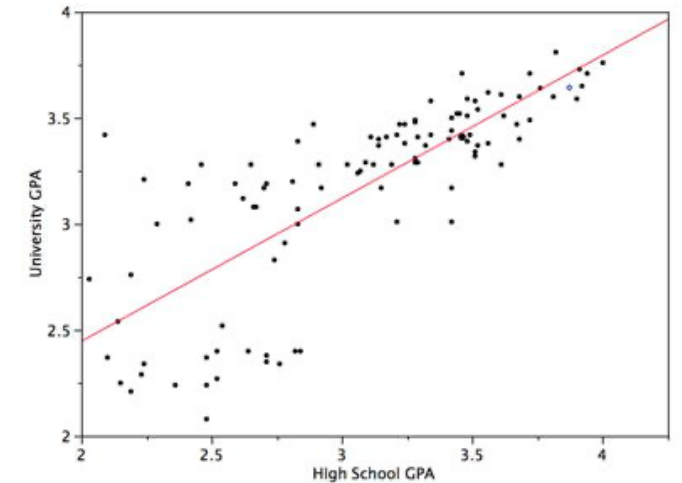
Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier 
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary

Linear Regression vs. Linear Classifier

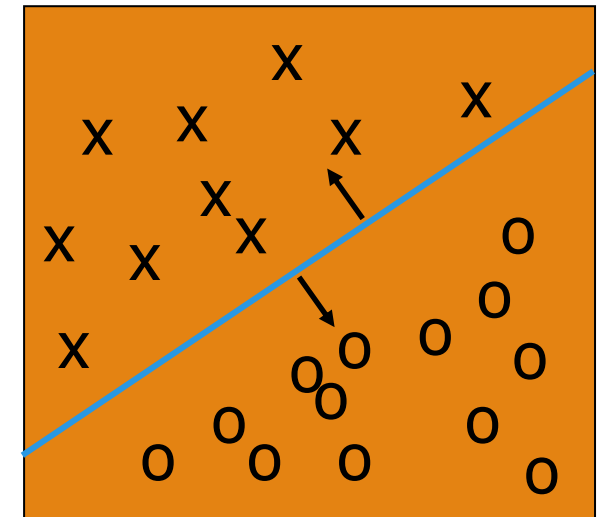
□ Linear regression

- Data modeled to fit a straight line
 - *Linear equation: $Y = wX + b$*
- Often uses the least-square method to fit the line
- Used to predict continuous values



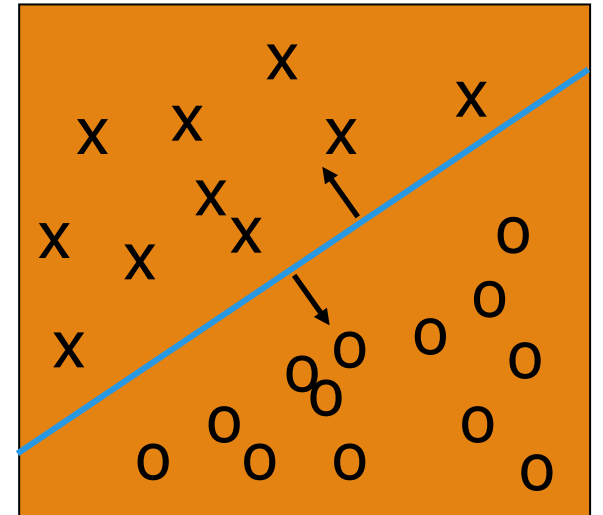
□ Linear Classifier

- Built a classification model using a straight line
- Used for (categorical data) binary classification



Linear Classifier: General Ideas

- ❑ Binary Classification
- ❑ $f(x)$ is a linear function based on the example's attribute values
 - ❑ The prediction is based on the value of $f(x)$
 - ❑ Data above the blue line belongs to class 'x' (i.e., $f(x) > 0$)
 - ❑ Data below blue line belongs to class 'o' (i.e., $f(x) < 0$)
- ❑ Classical Linear Classifiers
 - ❑ Linear Discriminant Analysis (LDA) (not covered)
 - ❑ Logistic Regression
 - ❑ Perceptron (later)
 - ❑ SVM (later)

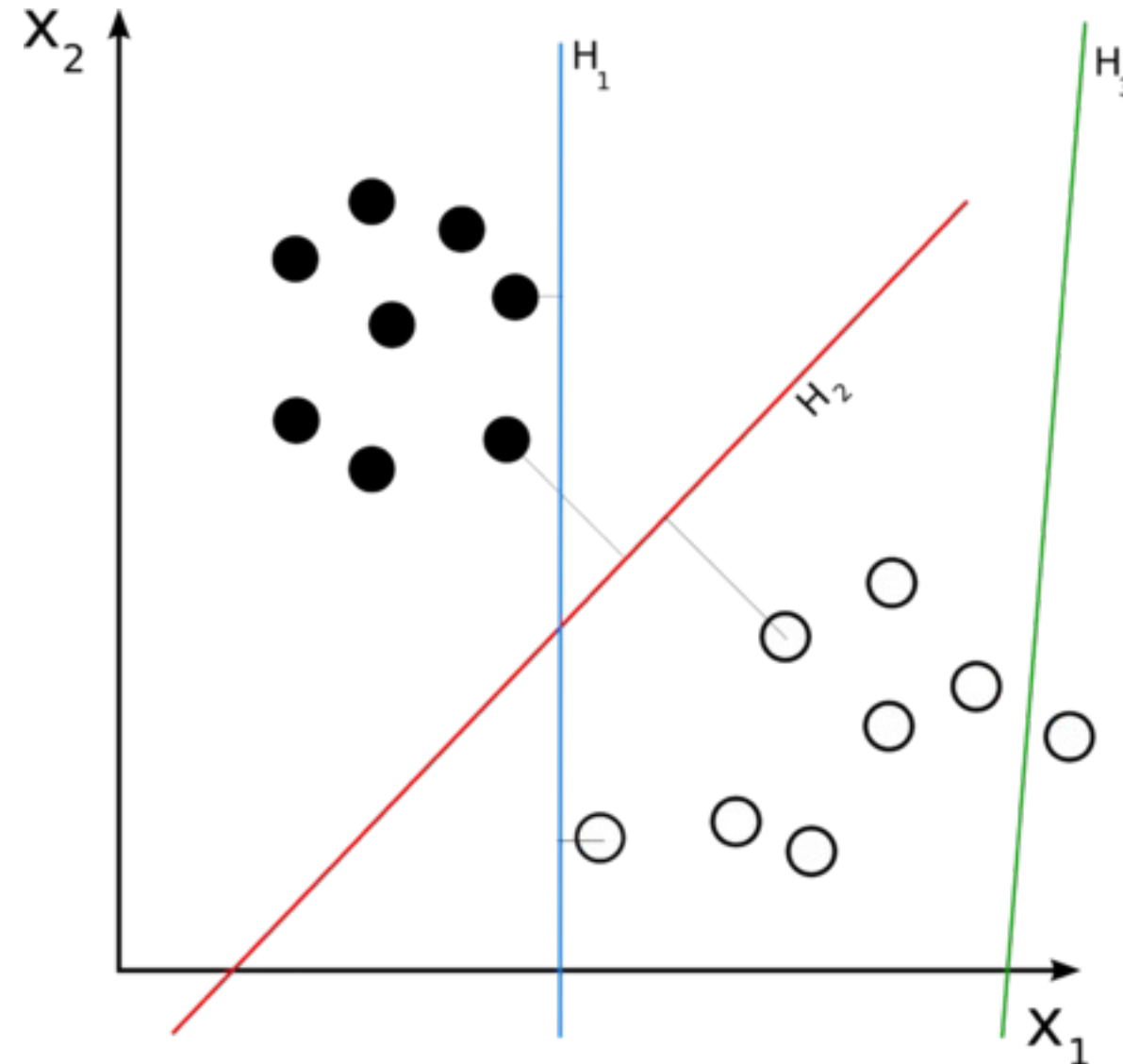


Linear Classifier: An Example

- A toy rule to determine whether a faculty member has tenure
 - $\text{Year} \geq 6 \text{ or Title} = \text{"Professor"} \Leftrightarrow \text{Tenure}$
- How to express the rule as a linear classifier?
- Features
 - $x_1 (x_1 \geq 0)$ is an integer denoting the year
 - x_2 is a Boolean denoting whether the title is "Professor"
- A feasible linear classifier: $f(x) = (x_1 - 5) + 6 \cdot x_2$
 - When x_2 is True, because $x_1 \geq 0$, $f(x)$ is always greater than 0
 - When x_2 is False, because $f(x) > 0 \Leftrightarrow x_1 \geq 6$
- There are many more feasible classifiers
 - $f(x) = (x_1 - 5.5) + 6 \cdot x_2$
 - $f(x) = 2 \cdot (x_1 - 5) + 11 \cdot x_2$
 -

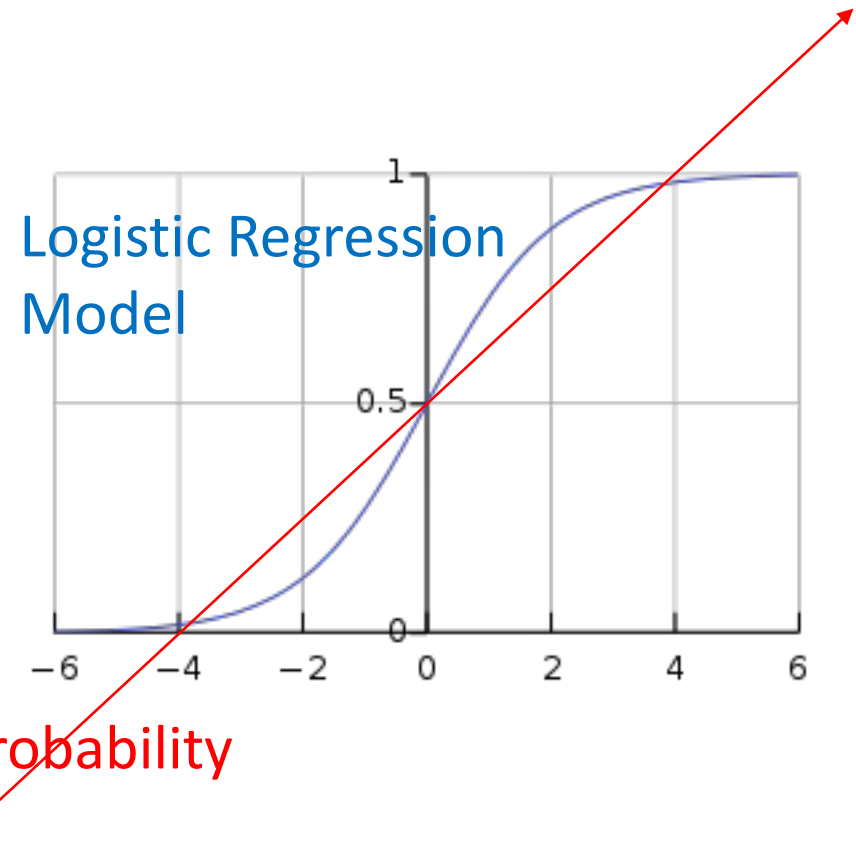
Key Question: Which Line Is Better?

- There might be many feasible linear functions
 - Both H_1 and H_2 will work
- Which one is better?
 - H_2 looks “better” in the sense that it is also furthest from both groups
 - We will introduce more in the SVM section



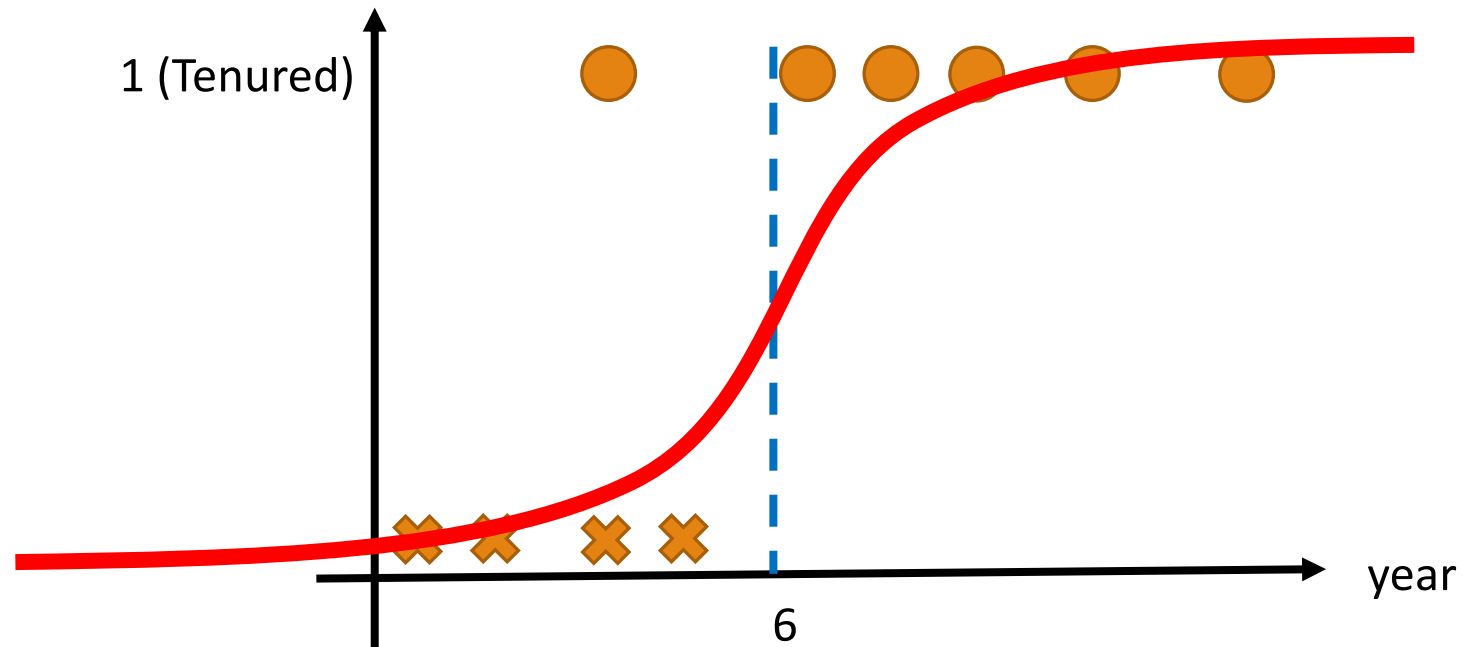
Logistic Regression: General Ideas

- Key Idea: Turns linear predictions into probabilities
- Sigmoid function:
 - $S(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{e^x+1}$
 - Projects $(-\infty, +\infty)$ to $[0, 1]$
- Compare to linear probability model
 - More smooth



Logistic Regression: An Example

- Suppose we only consider the year as feature



Logistic Regression: Maximum Likelihood

- The prediction function to learn

- $p(Y = 1 | X = x; \mathbf{w}) = S(w_0 + \sum_{i=1}^n w_i \cdot x_i)$

- $\mathbf{w} = (w_0, w_1, w_2, \dots, w_n)$ are the parameters

- Maximum Likelihood

- Log likelihood:

$$l(\mathbf{w}) = \sum_{i=1}^N y_i \log p(Y = 1 | X = x_i; \mathbf{w}) + (1 - y_i) \log(1 - p(Y = 1 | X = x_i; \mathbf{w}))$$

- There's no close form solution

- Gradient Descent

- Update $\mathbf{w}$ based on training data

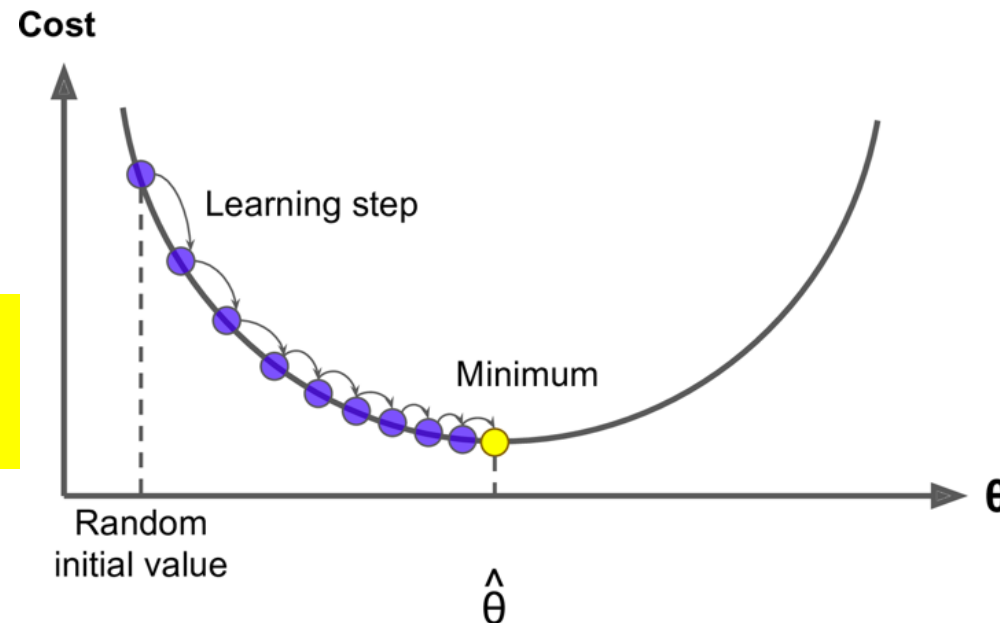
- Chain-rule for the gradient

Gradient Descent

- ❑ Gradient Descent is an iterative optimization algorithm for finding the minimum of a function (e.g., the negative log likelihood)
- ❑ For a function $F(x)$ at a point $\mathbf{a}$, $F(x)$ *decreases fastest* if we go in the direction of the negative gradient of $\mathbf{a}$

$$\mathbf{a}_{n+1} = \mathbf{a}_n - \gamma \nabla F(\mathbf{a}_n)$$

When the gradient is zero, we arrive at the local minimum



Generative vs. Discriminative Classifiers

- ❑ X: observed variables (features)
- ❑ Y: target variables (class labels)
- ❑ A generative classifier models $p(Y, X)$
 - ❑ It models how the data was "generated"? "what is the likelihood this or that class generated this instance?" and pick the one with higher probability
 - ❑ Naïve Bayes
 - ❑ Bayesian Networks
- ❑ A discriminative classifier models $p(Y | X)$
 - ❑ It uses the data to create a decision boundary
 - ❑ Logistic Regression
 - ❑ Support Vector Machines

Further Comments on Discriminative Classifiers

❑ Strength

- ❑ Prediction accuracy is generally high
 - ❑ As compared to generative models
 - ❑ Robust, works when training examples contain errors
- ❑ Fast evaluation of the learned target function
 - ❑ Comparing to (covered in future) Bayesian networks (which are normally slow)

❑ Criticism

- ❑ Long training time
- ❑ Difficult to understand the learned function (weights)
 - ❑ Bayesian networks can be used easily for pattern discovery
- ❑ Not easy to incorporate domain knowledge
 - ❑ Easy in the form of priors on the data or distributions

Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection 
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary

Model Evaluation and Selection

- ❑ Evaluation metrics
 - ❑ How can we measure accuracy?
 - ❑ Other metrics to consider?
- ❑ Use **validation test set** of class-labeled tuples instead of training set when assessing accuracy
- ❑ Methods for estimating a classifier's accuracy
 - ❑ Holdout method
 - ❑ Cross-validation
 - ❑ Bootstrap
- ❑ Comparing classifiers:
 - ❑ ROC Curves

Classifier Evaluation Metrics: Confusion Matrix

❑ Confusion Matrix:

Actual class \ Predicted class	$\oplus$ C_1	$\ominus$ $\neg C_1$
$\oplus$ C_1	True Positives (TP)	False Negatives (FN)
$\ominus$ $\neg C_1$	False Positives (FP)	True Negatives (TN)

❑ In a confusion matrix w. m classes, $CM_{i,j}$ indicates # of tuples in class i that were labeled by the classifier as class j

❑ May have extra rows/columns to provide totals

❑ Example of Confusion Matrix:

Actual class \ Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

Classifier Evaluation Metrics: Accuracy, Error Rate, Sensitivity and Specificity

A\P	C	¬C	
C	TP	FN	P
¬C	FP	TN	N
	P'	N'	All

- Classifier accuracy, or recognition rate
 - Percentage of test set tuples that are correctly classified

$$\text{Accuracy} = (TP + TN) / \text{All}$$

- Error rate: $1 - \text{accuracy}$, or
$$\text{Error rate} = (FP + FN) / \text{All}$$

- Class imbalance problem
 - One class may be *rare*
 - E.g., fraud, or HIV-positive
 - Significant *majority of the negative class* and minority of the positive class
 - Measures handle the class imbalance problem
 - Sensitivity** (recall): True positive recognition rate
 - $\text{Sensitivity} = TP / P$
 - Specificity**: True negative recognition rate
 - $\text{Specificity} = TN / N$

$$\frac{\text{คนที่ใช่ในทั้งหมด}}{\text{คนที่ใช่ในโรคทั้งหมด}}$$

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision:** Exactness: what % of tuples that the classifier labeled as positive are actually positive?

$$P = \text{Precision} = \frac{TP}{TP + FP}$$

ส่วน positive จริง ๆ ทั้งหมด

- **Recall:** Completeness: what % of positive tuples did the classifier label as positive?

$$R = \text{Recall} = \frac{TP}{TP + FN}$$

จำนวน tuple ที่ classifier จัดว่าเป็น positive
จำนวน tuple ที่เป็น positive ทั้งหมด } ex.

- Range: [0, 1]
- The “inverse” relationship between precision & recall
- **F measure (or F-score):** harmonic mean of precision and recall
- In general, it is the weighted measure of precision & recall

สิ่งนี้ทำให้มันได้ ทั้ง precision และ recall

$$F_{\beta} = \frac{1}{\alpha \cdot \frac{1}{P} + (1 - \alpha) \cdot \frac{1}{R}} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

Assigning β times as much weight to recall as to precision)

- **F1-measure (balanced F-measure)**

- That is, when $\beta = 1$,
$$F_1 = \frac{2PR}{P + R}$$

Classifier Evaluation Metrics: Example

- Use the same confusion matrix, calculate the measure just introduced

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	90	210	300	30.00 (<i>sensitivity</i>)
cancer = no	140	9560	9700	98.56 (<i>specificity</i>)
Total	230	9770	10000	96.50 (<i>accuracy</i>)

- Sensitivity = $TP/P = 90/300 = 30\%$
- Specificity = $TN/N = 9560/9700 = 98.56\%$
- Accuracy = $(TP + TN)/All = (90+9560)/10000 = 96.50\%$
- Error rate = $(FP + FN)/All = (140 + 210)/10000 = 3.50\%$
- Precision = $TP/(TP + FP) = 90/(90 + 140) = 90/230 = 39.13\%$
- Recall = $TP/ (TP + FN) = 90/(90 + 210) = 90/300 = 30.00\%$
- F1 = $2 P \times R / (P + R) = 2 \times 39.13\% \times 30.00\% / (39.13\% + 30\%) = 33.96\%$

Classifier Evaluation: Holdout & Cross-Validation

□ Holdout method

- Given data is randomly partitioned into two independent sets
 - Training set (e.g., 2/3) for model construction
 - Test set (e.g., 1/3) for accuracy estimation
- Repeated random sub-sampling validation: a variation of holdout
 - Repeat holdout k times, accuracy = avg. of the accuracies obtained

□ Cross-validation (k -fold, where $k = 10$ is most popular)

- Randomly partition the data into k *mutually exclusive* subsets, each approximately equal size
- At i -th iteration, use D_i as test set and others as training set
- Leave-one-out: k folds where $k = \#$ of tuples, for small sized data
- *Stratified cross-validation*: folds are stratified so that class distribution, in each fold is approximately the same as that in the initial data

Classifier Evaluation: Bootstrap

□ Bootstrap

- Works well with small data sets
- Samples the given training tuples uniformly *with replacement*
 - Each time a tuple is selected, it is equally likely to be selected again and re-added to the training set

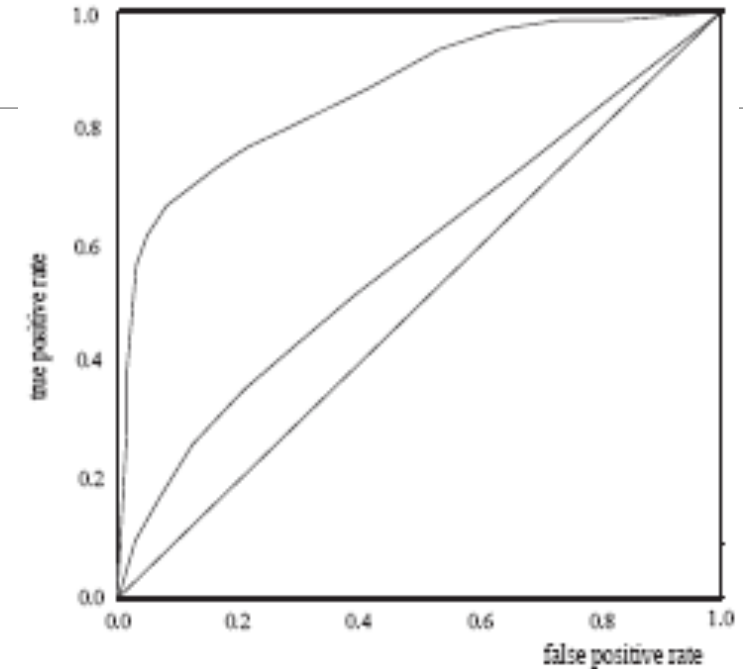
□ Several bootstrap methods, and a common one is **.632 bootstrap**

- A data set with d tuples is sampled d times, with replacement, resulting in a training set of d samples. The data tuples that did not make it into the training set end up forming the test set. About 63.2% of the original data end up in the bootstrap, and the remaining 36.8% form the test set (since $(1 - 1/d)^d \approx e^{-1} = 0.368$)
- Repeat the sampling procedure k times, overall accuracy of the model:

$$Acc(M) = \frac{1}{k} \sum_{i=1}^k (0.632 \times Acc(M_i)_{test_set} + 0.368 \times Acc(M_i)_{train_set})$$

Model Selection: ROC Curves

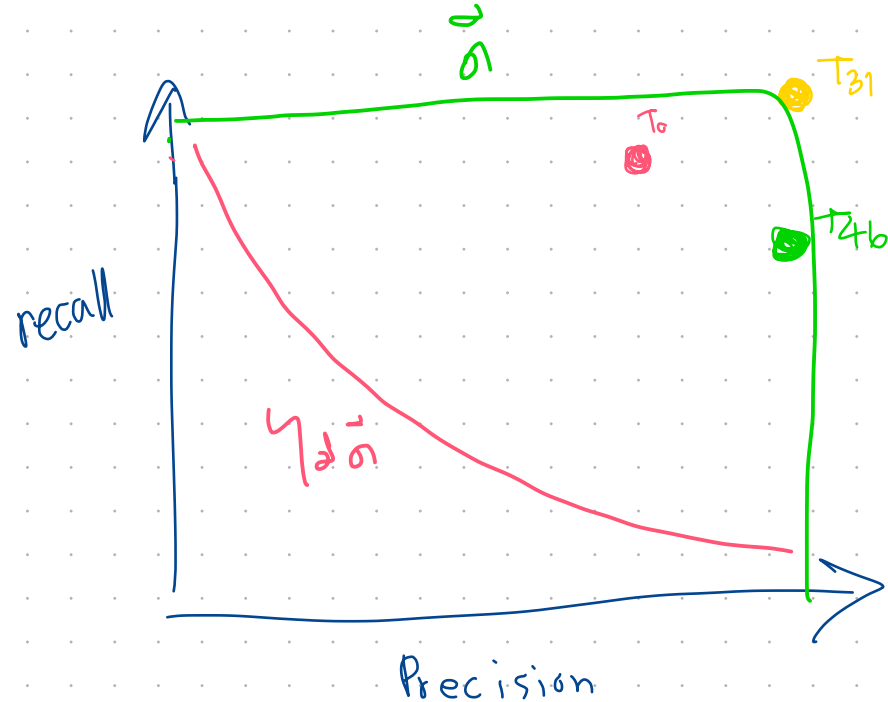
- ❑ **ROC** (Receiver Operating Characteristics) curves: for visual comparison of classification models
- ❑ Originated from signal detection theory
- ❑ Shows the trade-off between the true positive rate and the false positive rate
- ❑ The area under the ROC curve (**AUC**: Area Under Curve) is a measure of the accuracy of the model
- ❑ Rank the test tuples in decreasing order: the one that is most likely to belong to the positive class appears at the top of the list
- ❑ The closer to the diagonal line (i.e., the closer the area is to 0.5), the less accurate is the model



- ❑ Vertical axis represents the true positive rate
- ❑ Horizontal axis rep. the false positive rate
- ❑ The plot also shows a diagonal line
- ❑ A model with perfect accuracy will have an area of 1.0

DT →

Positive	Prob.	67 %	P	T_6	T_{31}	T_{46}
Positive	Prob.	30 %	N	P	P	P
Positive	Prob.	90 %	P	P	N	N
Positive	Prob.	45 %	P	P	P	P
				R_{100}	R_{75}	R_{100}
					R_{100}	R_{100}
					R_{66}	R_{100}



Issues Affecting Model Selection

- ❑ **Accuracy**

- ❑ classifier accuracy: predicting class label

- ❑ **Speed**

- ❑ time to construct the model (training time)
 - ❑ time to use the model (classification/prediction time)

- ❑ **Robustness:** handling noise and missing values

- ❑ **Scalability:** efficiency in disk-resident databases

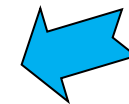
- ❑ **Interpretability**

- ❑ understanding and insight provided by the model

- ❑ Other measures, e.g., goodness of rules, such as decision tree size or compactness of classification rules

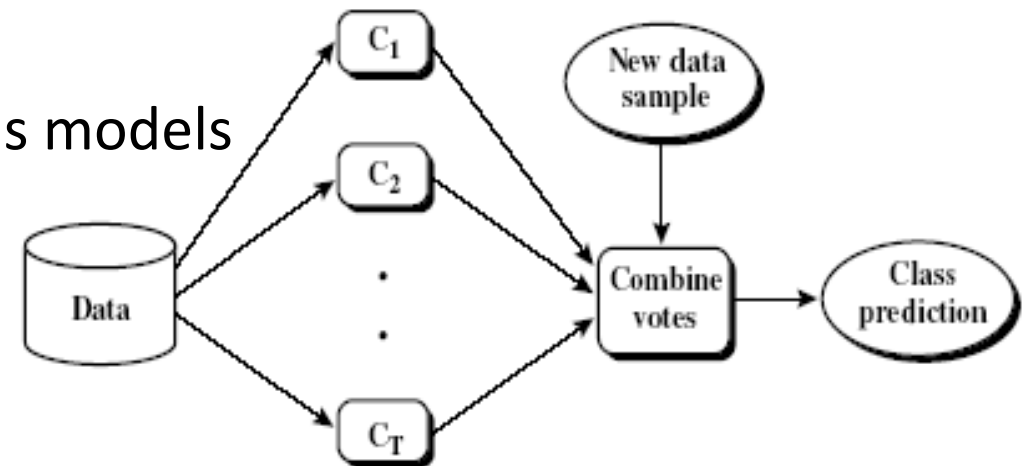
Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary



Ensemble Methods: Increasing the Accuracy

- ❑ Ensemble methods
 - ❑ Use a combination of models to increase accuracy
 - ❑ Combine a series of k learned models, $M_1, M_2, \dots, M_k$, with the aim of creating an improved model M^*
- ❑ Popular ensemble methods
 - ❑ Bagging: Trains each model using a subset of the training set, and models learned in parallel
 - ❑ Boosting: Trains each new model instance to emphasize the training instances that previous models mis-classified, and models learned in order



Bagging: Bootstrap Aggregation

- ❑ Analogy: Diagnosis based on multiple doctors' majority vote
- ❑ Training
 - ❑ Given a set D of d tuples, at each iteration i , a training set D_i of d tuples is sampled with replacement from D (i.e., bootstrap)
 - ❑ A classifier model M_i is learned for each training set D_i
- ❑ Classification: classify an unknown sample X
 - ❑ Each classifier M_i returns its class prediction
 - ❑ The bagged classifier M^* counts the votes and assigns the class with the most votes to X
- ❑ Prediction: It can be applied to the prediction of continuous values by taking the average value of each prediction for a given test tuple
- ❑ Accuracy: Improved accuracy in prediction
 - ❑ Often significantly better than a single classifier derived from D
 - ❑ For noise data: Not considerably worse, more robust

Random Forest: Basic Concepts

- ❑ Random Forest (first proposed by L. Breiman in 2001)
 - ❑ A variation of bagging for *decision trees*
 - ❑ *Data bagging*
 - ❑ Use a subset of training data by sampling with replacement for each tree
 - ❑ *Feature bagging*
 - ❑ At each node use a random selection of attributes as candidates and split by the best attribute among them
 - ❑ Compared to original bagging, increases the diversity among generated trees
 - ❑ During classification, each tree votes and the most popular class is returned

Random Forest

- ❑ Two Methods to construct Random Forest:
 - ❑ Forest-RI (*random input selection*): Randomly select, at each node, F attributes as candidates for the split at the node. The CART methodology is used to grow the trees to maximum size
 - ❑ Forest-RC (*random linear combinations*): Creates new attributes (or features) that are a linear combination of the existing attributes (reduces the correlation between individual classifiers)
- ❑ Comparable in accuracy to Adaboost, but more robust to errors and outliers
- ❑ Insensitive to the number of attributes selected for consideration at each split, and faster than typical bagging or boosting

Boosting

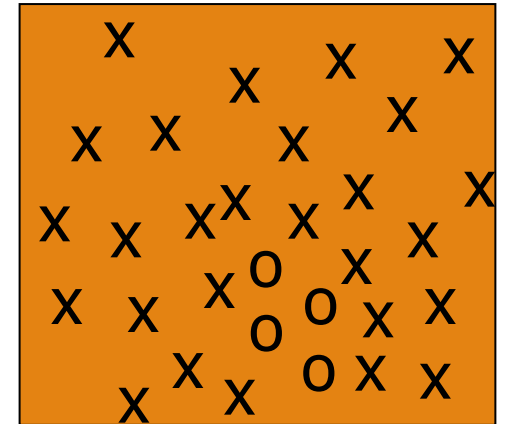
- ❑ Analogy: Consult several doctors, based on a combination of weighted diagnoses—weight assigned based on the previous diagnosis accuracy
- ❑ How boosting works?
 - ❑ **Weights** are assigned to each training tuple
 - ❑ A series of k classifiers is iteratively learned
 - ❑ After a classifier M_i is learned, the weights are updated to allow the subsequent classifier, M_{i+1} , to **pay more attention to the training tuples that were misclassified** by M_i
 - ❑ The final **M^* combines the votes** of each individual classifier, where the weight of each classifier's vote is a function of its accuracy
- ❑ Boosting algorithm can be extended for numeric prediction
- ❑ Comparing with bagging: Boosting tends to have greater accuracy, but it also risks overfitting the model to misclassified data

Adaboost (Freund and Schapire, 1997)

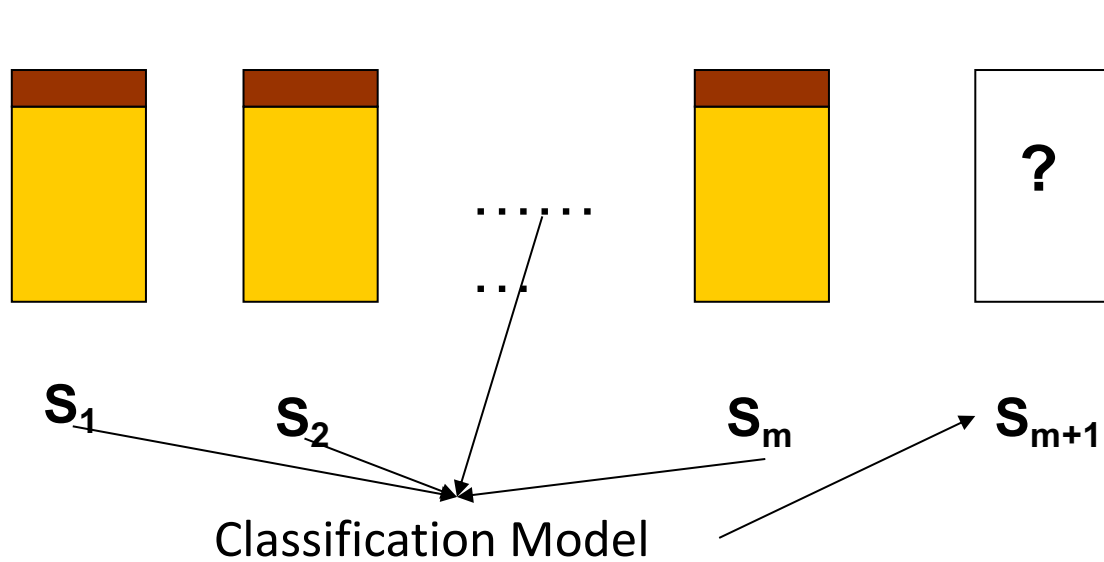
- ❑ Given a set of d class-labeled tuples, $(\mathbf{X}_1, y_1), \dots, (\mathbf{X}_d, y_d)$
- ❑ Initially, all the weights of tuples are set the same $(1/d)$
- ❑ Generate k classifiers in k rounds. At round i ,
 - ❑ Tuples from D are sampled (with replacement) to form a training set D_i of the same size
 - ❑ Each tuple's chance of being selected is based on its weight
 - ❑ A classification model M_i is derived from D_i
 - ❑ Its error rate is calculated using D_i as a test set
 - ❑ If a tuple is misclassified, its weight is increased; otherwise, it is decreased
- ❑ Error rate: $err(\mathbf{X}_j)$ is the misclassification error of tuple $\mathbf{X}_j$. Classifier M_i error rate is the sum of the weights of the misclassified tuples:
$$error(M_i) = \sum_j^d w_j \times err(\mathbf{X}_j)$$
- ❑ The weight of classifier M_i 's vote is $\log \frac{1 - error(M_i)}{error(M_i)}$

Classification of Class-Imbalanced Data Sets

- ❑ Class-imbalance problem: Rare positive examples but numerous negative ones
 - ❑ E.g., medical diagnosis, fraud transaction, accident (oil-spill), and product fault
- ❑ Traditional methods assume a balanced distribution of classes and equal error costs: not suitable for class-imbalanced data
- ❑ Typical methods on imbalanced data in two-class classification
 - ❑ **Oversampling:** Re-sampling of data from positive class
 - ❑ **Under-sampling:** Randomly eliminate tuples from negative class
 - ❑ **Threshold-moving:** Move the decision threshold, t , so that the rare class tuples are easier to classify, and hence, less chance of costly false negative errors
 - ❑ **Ensemble techniques:** Ensemble multiple classifiers introduced above
- ❑ Still difficult for class imbalance problem on multiclass tasks



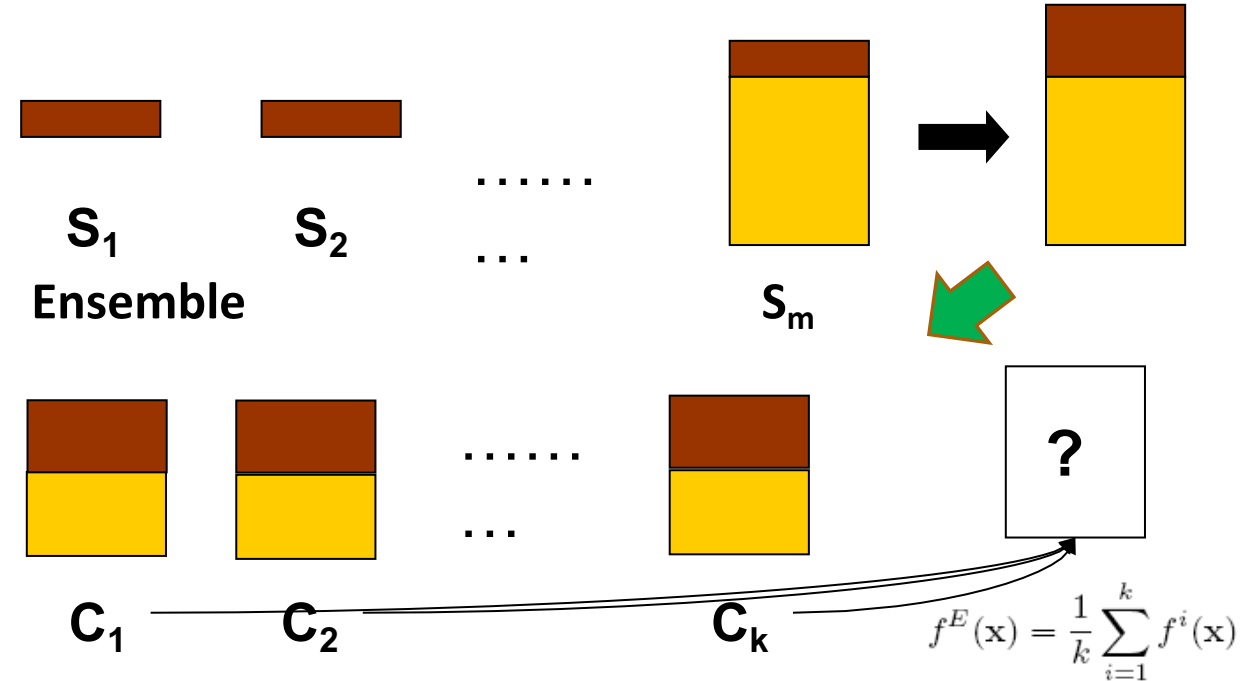
Classifying Data Streams with Skewed Distribution



S_m as training data? Two few positive examples!

J. Gao, et al., "A General Framework for Mining Concept-Drifting Data Streams with Skewed Distributions", SDM'07

Biased Sampling



- Classify data stream with skewed distribution (i.e., rare events)
 - **Biased sampling:** Save only the positive examples in the streams
 - **Ensemble:** Partition negative examples of S_m into k portions to build k classifiers
 - Effectively reduce classification errors on the minority class

Chapter 8. Classification: Basic Concepts

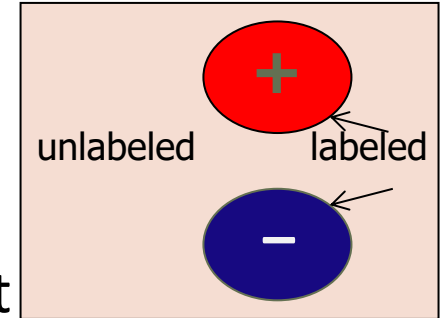
- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification 
- ❑ Summary

Multiclass Classification

- ❑ Classification involving more than two classes (i.e., > 2 Classes)
- ❑ Methodology: Reducing the multi-class problem into multiple binary problems
- ❑ Method 1. **One-vs.-rest** (or **one-vs.-all**)
 - ❑ Given m classes, train m classifiers: one for each class
 - ❑ Classifier j : treat tuples in class j as *positive* & **all the rest** as *negative*
 - ❑ To classify a tuple $\mathbf{X}$, the set of classifiers vote as an ensemble
- ❑ Method 2. **one-vs.-one** (or **all-vs.-all**): Learn a classifier for each pair of classes
 - ❑ Given m classes, construct $m(m - 1)/2$ binary classifiers
 - ❑ A classifier is trained using tuples of the two classes
 - ❑ To classify a tuple $\mathbf{X}$, each classifier votes
 - ❑ $\mathbf{X}$ is assigned to the class with maximal vote
- ❑ Comparison: One-vs.-one tends to perform better than one-vs.-rest
- ❑ Many new algorithms have been developed to go beyond binary classifier method

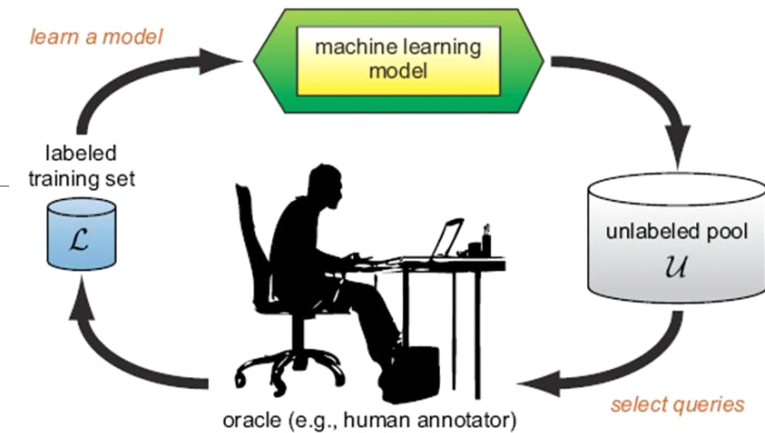
Semi-Supervised Classification

- ❑ Semi-supervised: Uses labeled and unlabeled data to build a classifier
- ❑ Self-training
 - ❑ Build a classifier using the labeled data
 - ❑ Use it to label the unlabeled data, and those with the most confident label prediction are added to the set of labeled data
 - ❑ Repeat the above process
 - ❑ Adv.: easy to understand; Disadv.: may reinforce errors
- ❑ Co-training: Use two or more classifiers to teach each other
 - ❑ Each learner uses a mutually independent set of features of each tuple to train a good classifier, say f_1 and f_2
 - ❑ Then f_1 and f_2 are used to predict the class label for unlabeled data X
 - ❑ Teach each other: The tuple having the most confident prediction from f_1 is added to the set of labeled data for f_2 & vice versa
- ❑ Other methods include joint probability distribution of features and labels



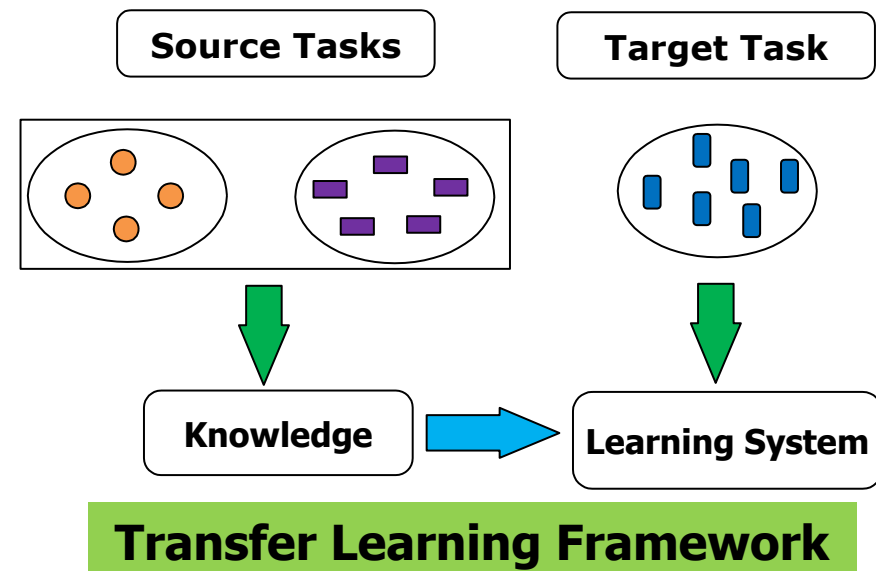
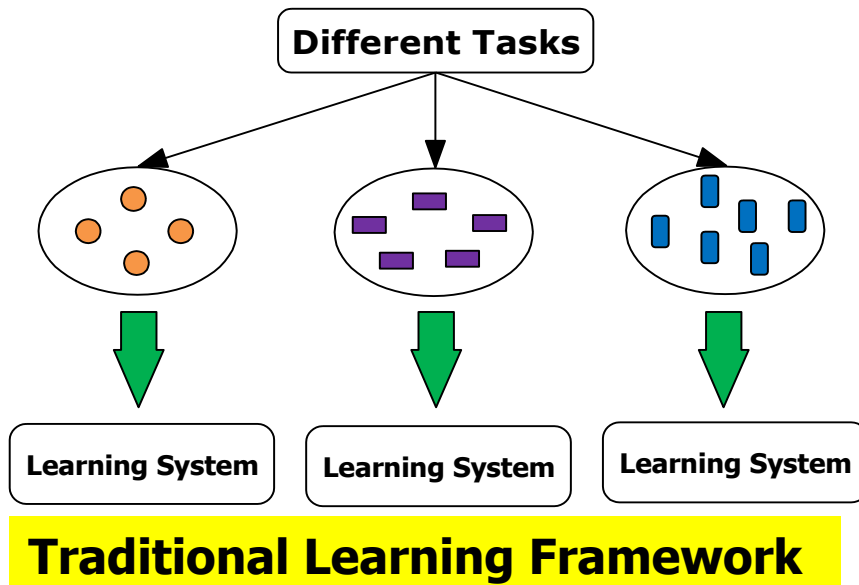
Active Learning

- A special case of semi-supervised learning
 - Unlabeled data: Abundant
 - Class labels are expensive to obtain
- Active learner: Interactively query teachers (oracle) for labels
- Pool-based approach: Uses a pool of unlabeled data
 - L : a small subset of D is labeled, U : a pool of unlabeled data in D
 - Use a query function to carefully select one or more tuples from U and request labels from an oracle (a human annotator)
 - The newly labeled samples are added to L , and learn a model
 - Goal: **Achieve high accuracy using as few labeled data as possible**
- Evaluated using *learning curves*: Accuracy as a function of the number of instances queried (# of tuples to be queried should be small)
- A lot of algorithms have been developed for active learning



Transfer Learning: Conceptual Framework

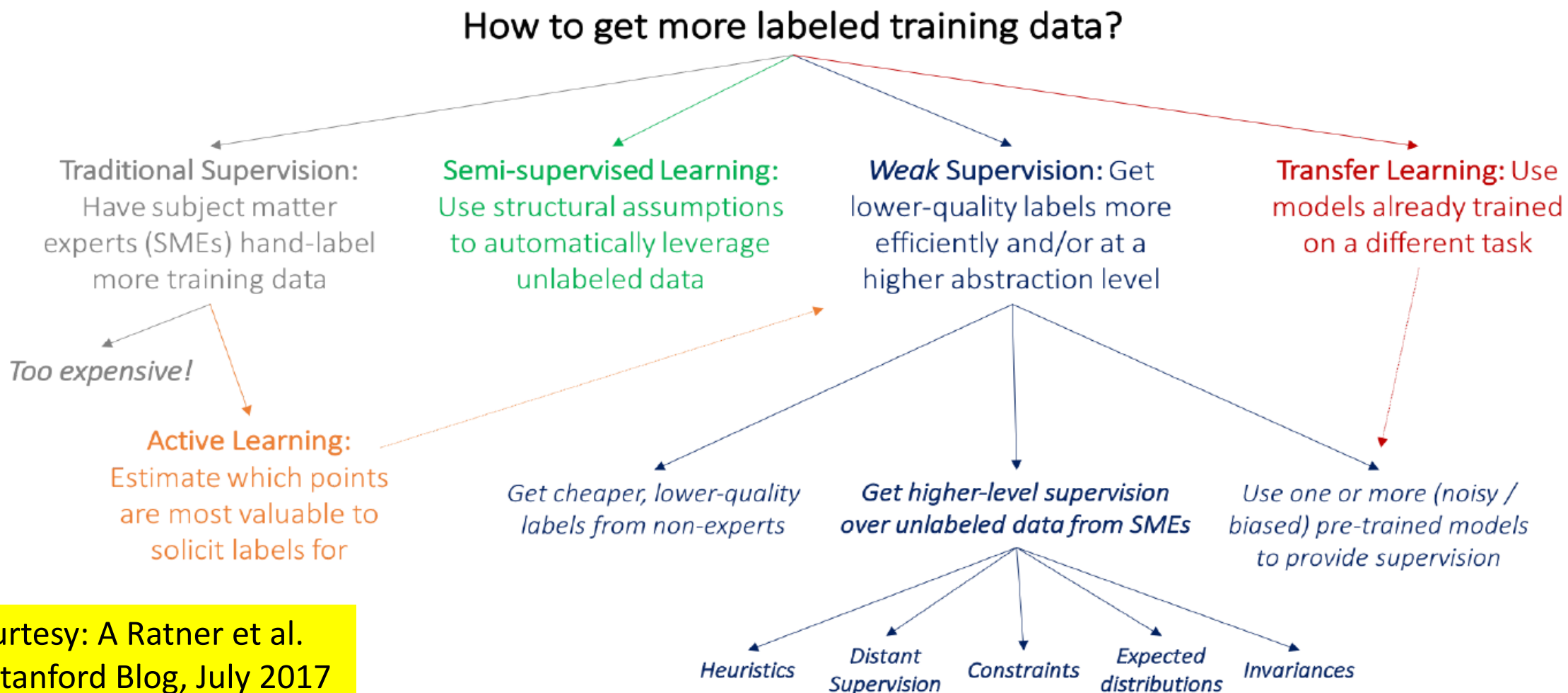
- ❑ Transfer learning: Extract knowledge from one or more source tasks (e.g., recognizing cars) and apply the knowledge to a target task (e.g., recognizing trucks)
- ❑ Traditional learning: Build a new classifier for each new task
- ❑ Transfer learning: Build new classifier by applying existing knowledge learned from source tasks
- ❑ Many algorithms are developed, applied to text classification, spam filtering, etc.



Weak Supervision: A New Programming Paradigm for Machine Learning

- ❑ Overcome the training data bottleneck
 - ❑ Leverage higher-level and/or noisier input from experts
- ❑ Exploring weak label distributions provided more cheaply and efficiently by
 - ❑ Higher-level, less precise supervision (e.g., heuristic rules, expected label distributions)
 - ❑ Cheaper, lower-quality supervision (e.g. crowdsourcing)
 - ❑ Existing resources (e.g. knowledge bases, pre-trained models)
- ❑ These weak label distributions could take many forms
 - ❑ Weak Labels from crowd workers, output of heuristic rules, or the result of distant supervision (from KBs), or the output of other classifiers, etc.
 - ❑ Constraints and invariances (e.g., from physics, logic, or other experts)
 - ❑ Probability distributions (e.g., from weak or biased classifiers or user-provided label or feature expectations or measurements)

Relationships Among Different Kinds of Supervisions



Courtesy: A Ratner et al.
@Stanford Blog, July 2017

Many areas of machine learning are motivated by the bottleneck of labeled training data, but are divided at a high-level by what information they leverage instead.

Chapter 8. Classification: Basic Concepts

- ❑ Classification: Basic Concepts
- ❑ Decision Tree Induction
- ❑ Bayes Classification Methods
- ❑ Linear Classifier
- ❑ Model Evaluation and Selection
- ❑ Techniques to Improve Classification Accuracy: Ensemble Methods
- ❑ Additional Concepts on Classification
- ❑ Summary 

Summary

- ❑ Classification: Model construction from a set of training data
- ❑ Effective and scalable methods
 - ❑ Decision tree induction, Bayes classification methods, linear classifier, ...
 - ❑ No single method has been found to be superior over all others for all data sets
- ❑ Evaluation metrics: Accuracy, sensitivity, specificity, precision, recall, F measure
- ❑ Model evaluation: Holdout, cross-validation, bootstrapping, ROC curves (AUC)
- ❑ Improve Classification Accuracy: Bagging, boosting
- ❑ Additional concepts on classification: Multiclass classification, semi-supervised classification, active learning, transfer learning, weak supervision

References (1)

- ❑ C. Apte and S. Weiss. **Data mining with decision trees and decision rules**. Future Generation Computer Systems, 13, 1997
- ❑ P. K. Chan and S. J. Stolfo. **Learning arbiter and combiner trees from partitioned data for scaling machine learning**. KDD'95
- ❑ A. J. Dobson. **An Introduction to Generalized Linear Models**. Chapman & Hall, 1990.
- ❑ R. O. Duda, P. E. Hart, and D. G. Stork. **Pattern Classification**, 2ed. John Wiley, 2001
- ❑ U. M. Fayyad. **Branching on attribute values in decision tree generation**. AAAI'94.
- ❑ Y. Freund and R. E. Schapire. **A decision-theoretic generalization of on-line learning and an application to boosting**. J. Computer and System Sciences, 1997.
- ❑ J. Gehrke, R. Ramakrishnan, and V. Ganti. **Rainforest: A framework for fast decision tree construction of large datasets**. VLDB'98.
- ❑ J. Gehrke, V. Gant, R. Ramakrishnan, and W.-Y. Loh, **BOAT -- Optimistic Decision Tree Construction**. SIGMOD'99.
- ❑ T. Hastie, R. Tibshirani, and J. Friedman. **The Elements of Statistical Learning: Data Mining, Inference, and Prediction**. Springer-Verlag, 2001.

References (2)

- ❑ T.-S. Lim, W.-Y. Loh, and Y.-S. Shih. **A comparison of prediction accuracy, complexity, and training time of thirty-three old and new classification algorithms.** Machine Learning, 2000
- ❑ J. Magidson. **The Chaid approach to segmentation modeling: Chi-squared automatic interaction detection.** In R. P. Bagozzi, editor, Advanced Methods of Marketing Research, Blackwell Business, 1994
- ❑ M. Mehta, R. Agrawal, and J. Rissanen. **SLIQ : A fast scalable classifier for data mining.** EDBT'96
- ❑ T. M. Mitchell. **Machine Learning.** McGraw Hill, 1997
- ❑ S. K. Murthy, **Automatic Construction of Decision Trees from Data: A Multi-Disciplinary Survey,** Data Mining and Knowledge Discovery 2(4): 345-389, 1998
- ❑ J. R. Quinlan. **Induction of decision trees.** *Machine Learning*, 1:81-106, 1986.
- ❑ J. R. Quinlan. **C4.5: Programs for Machine Learning.** Morgan Kaufmann, 1993.
- ❑ J. R. Quinlan. **Bagging, boosting, and c4.5.** AAAI'96.

References (3)

- ❑ R. Rastogi and K. Shim. **Public: A decision tree classifier that integrates building and pruning.** VLDB'98
- ❑ J. Shafer, R. Agrawal, and M. Mehta. **SPRINT : A scalable parallel classifier for data mining.** VLDB'96
- ❑ J. W. Shavlik and T. G. Dietterich. **Readings in Machine Learning.** Morgan Kaufmann, 1990
- ❑ P. Tan, M. Steinbach, and A. Karim. **Introduction to Data Mining.** Addison Wesley, 2005
- ❑ S. M. Weiss and C. A. Kulikowski. **Computer Systems that Learn: Classification and Prediction Methods from Statistics, Neural Nets, Machine Learning, and Expert Systems.** Morgan Kaufman, 1991
- ❑ S. M. Weiss and N. Indurkha. **Predictive Data Mining.** Morgan Kaufmann, 1997
- ❑ I. H. Witten and E. Frank. **Data Mining: Practical Machine Learning Tools and Techniques,** 2ed. Morgan Kaufmann, 2005



Bayes' Theorem: Basics

- Total probability Theorem:
$$P(B) = \sum_{i=1}^M P(B|A_i)P(A_i)$$
- Bayes' Theorem:
$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$
 - Let $\mathbf{X}$ be a data sample (“evidence”): class label is unknown
 - Let H be a *hypothesis* that X belongs to class C
 - Classification is to determine $P(H|\mathbf{X})$, (i.e., *posteriori probability*): the probability that the hypothesis holds given the observed data sample $\mathbf{X}$
 - $P(H)$ (*prior probability*): the initial probability
 - E.g., $\mathbf{X}$ will buy computer, regardless of age, income, ...
 - $P(\mathbf{X})$: probability that sample data is observed
 - $P(\mathbf{X}|H)$ (likelihood): the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds
 - E.g., Given that $\mathbf{X}$ will buy computer, the prob. that X is 31..40, medium income

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$
- Suppose there are m classes $C_1, C_2, \dots, C_m$.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i | \mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i | \mathbf{X}) = \frac{P(\mathbf{X} | C_i)P(C_i)}{P(\mathbf{X})}$$

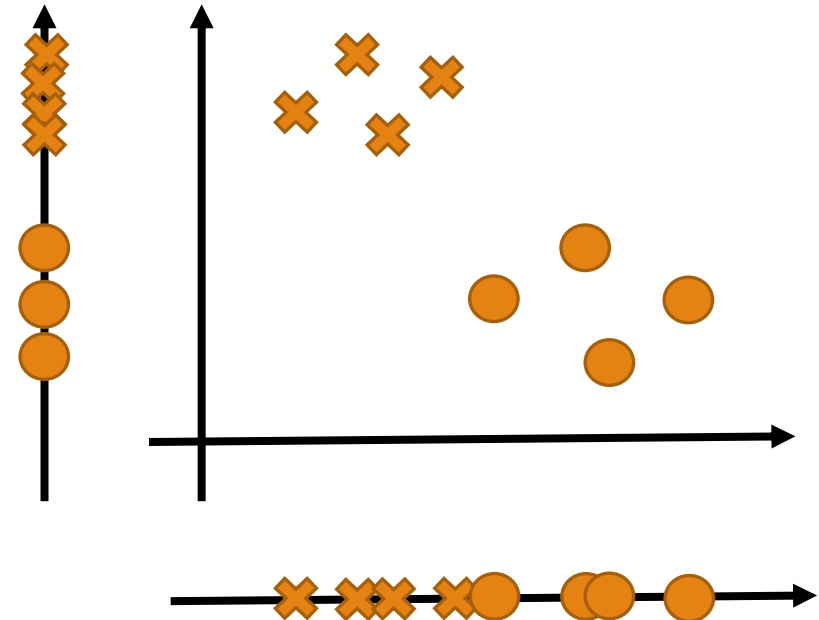
- Since $P(\mathbf{X})$ is constant for all classes, only

$$P(C_i | \mathbf{X}) \propto P(\mathbf{X} | C_i)P(C_i)$$

needs to be maximized

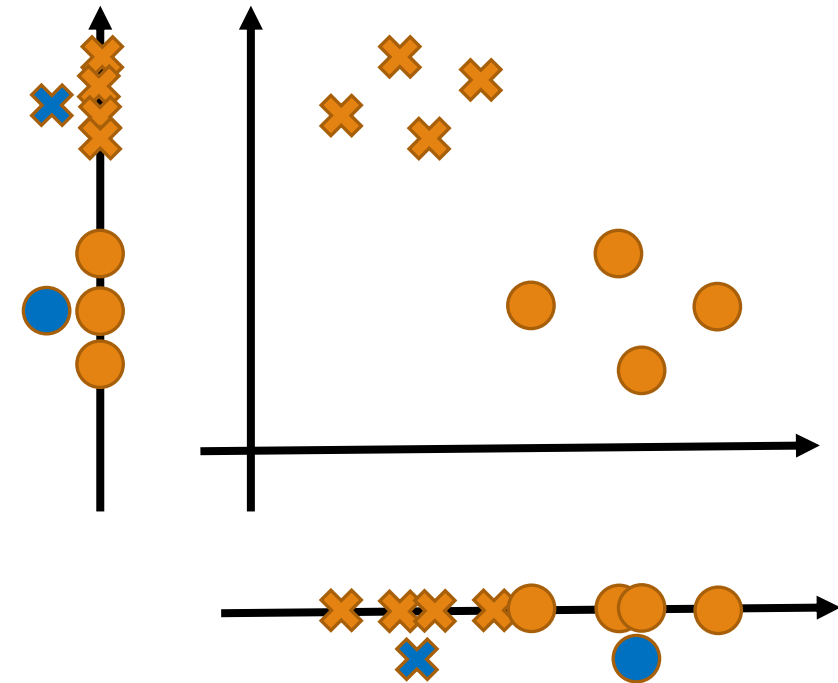
Linear Discriminant Analysis (LDA)

- Linear Discriminant Analysis (LDA) works when the attributes are all continuous
 - For the categorical attributes, discriminant correspondence analysis is the equivalent technique
- Basic Ideas: Project all samples on a line such that different classes are well separated
- Example: Suppose we have 2 classes and 2-dimensional samples $x_1, \dots, x_n$
 - n_1 samples come from class 1
 - n_2 samples come from class 2
- Let the line direction be given by unit vector $\mathbf{v}$
- There are two candidates of projections
 - Vertical: $\mathbf{v} = (0,1)$
 - Horizontal: $\mathbf{v} = (1,0)$
- Which one looks better?
- How to mathematically measure it?



Fisher's LDA (Linear Discriminant Analysis)

- $v^T x_i$ is the distance of projection of x_i from the origin
- Let μ_1 and μ_2 be the means of class 1 and class 2 in the original space
 - $\mu_1 = \frac{1}{n_1} \sum_{i \in \text{class 1}} x_i$
 - $\mu_2 = \frac{1}{n_2} \sum_{i \in \text{class 2}} x_i$
- The distance between the means of the projected points
 - $|v^T \mu_1 - v^T \mu_2|$
 - Good? No. Horizontal one may have larger distance



Fisher's LDA (con't)

- Normalization needed
- Scatter: Sample variance multiplied by n

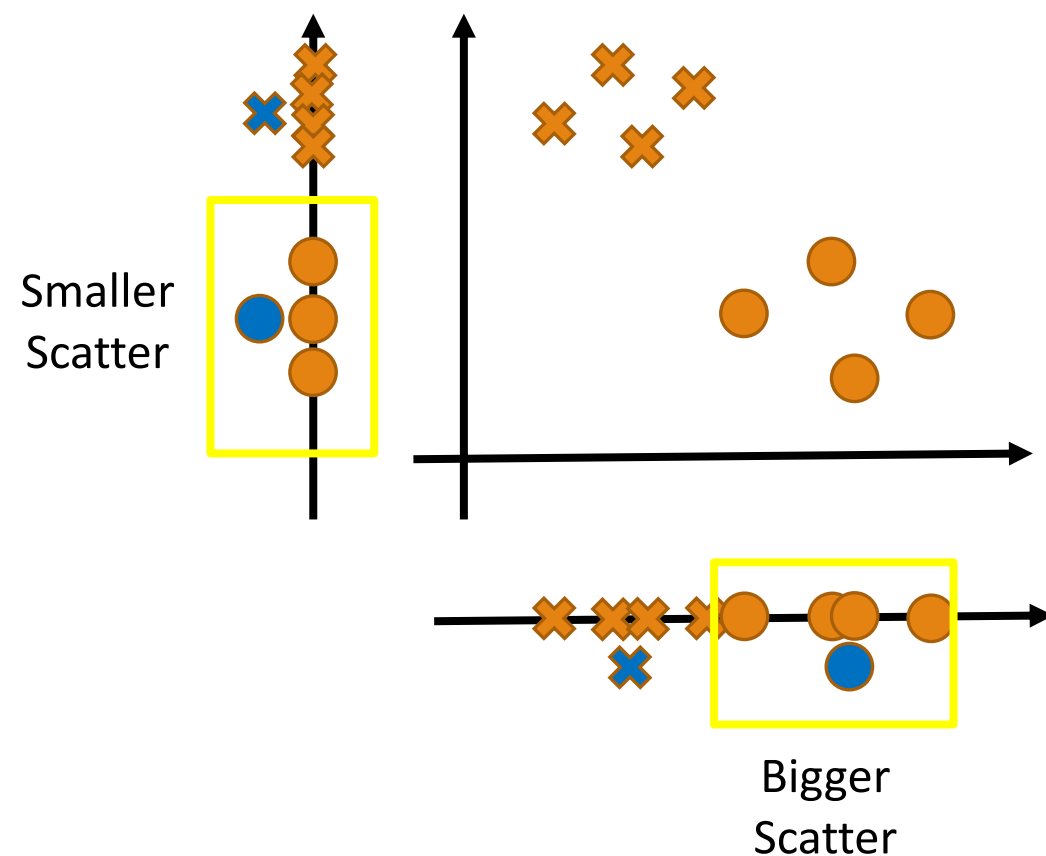
- $s_1 = \sum_{i \in \text{class } 1} (v^T x_i - v^T \mu_1)^2$

- $s_2 = \sum_{i \in \text{class } 2} (v^T x_i - v^T \mu_2)^2$

- Fisher's LDA

- Maximize $J(v) = \frac{(v^T \mu_1 - v^T \mu_2)^2}{s_1 + s_2}$

- Closed-form optimal solution



Fisher's LDA: Summary

- Advantages
 - Useful for dimension reduction
 - Easy to extend to multi-classes

- Fisher's LDA will fail
 - When $\mu_1 = \mu_2$, $J(\mathbf{v})$ is always 0.
 - When classes have large overlap when projected to any line